Meibomian Gland Segmentation on a Public Benchmark: Patient-Level Leakage, Tile-Size Confounders, and Uncertainty-Guided Review

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Abstract

Deep learning for infrared meibography segmentation has advanced rapidly, but the comparability of published results is limited by three neglected issues: image-level splits that let the same patient appear in training and test, unreported tiling parameters at inference, and a gap between pixel metrics and clinical Meiboscore. We establish a patient-independent, reproducible benchmark on the public MGD-1K dataset: 999 deduplicated pairs from 338 subjects, split by patient into 694/153/152 (239/50/49 patients), with the split file and assignment signature released.

We first quantify a hidden leakage: the official 200-case Meiboscore test list contains 166 patients, while 326 of the remaining 800 images come from the same patients; even a patient-level training set still contains 116/166 of the test patients. We then compare UNet, Attention U-Net, WRCANet, MGD-PriorNet, and MGD-Former under one preprocessing, one patient-level split, one loss, and one inference protocol; WRCANet did not converge within the 8 GB budget, so the main table ranks the four comparable routes and reports WRCANet separately as a feasibility counterexample. MGD-Former uses an ADE20K-pretrained MiT-B2 encoder and a stride-4 high-resolution fusion decoder.

We treat tile size as a first-class experimental variable: with identical weights, changing only tile height moves Dice from 0.370 to 0.655. This motivates Multi-Tile Ensemble (MTE), which averages probabilities from 256/384/512 tiles. MTE improved HD95 but not Dice. We further compare multi-tile variance, predictive entropy, and 8× TTA by AUROC for failure detection, and report agreement between segmentation-derived R_loss/Meiboscore and six-round expert grading, together with upper/lower eyelid subgroup differences. Clinical agreement is only QWK 0.131, showing that pixel metrics cannot replace expert grading; TTA8 AUROC is near 1.0 but suffers from label–signal coupling and is not direct triage evidence. All accuracy values come from unified training and a patient-level independent test; no unmeasured performance is reported.

Keywords: meibomian gland segmentation; public benchmark; patient-level leakage; tiling ensemble; uncertainty; clinical grading

1 Introduction

Meibomian gland dysfunction (MGD) is among the most common causes of dry eye disease[1, 2]. Infrared/near-infrared meibography visualises gland dropout, tortuosity, and shortening non-invasively, and grading systems such as Meiboscore map dropout to clinical severity[3, 4]. Manual grading is observer-dependent and only moderately repeatable. Pixel-wise segmentation of glands and eyelid margin is therefore a prerequisite for quantitative area, length, tortuosity, and R_loss.

This task differs from generic semantic segmentation in three respects. First, targets are extremely thin: at 640×1280, a gland is 3–8 pixels wide. Any stride-16 backbone reduces it below one pixel; fragmentation appears before Dice drops, and is exposed mainly by HD95 and skeleton continuity. Second, contrast is low and cross-device variation is large. Third, data are scarce: about one thousand paired cases cannot train a large Transformer from scratch, making pretraining and strong regularisation necessary[5, 6].

Public data have changed the field. MGD-1K provides 1000 LipiView II infrared images, gland and eyelid masks, demographics for 320 subjects, and six expert Meiboscore rounds per image[5]; recent multicentre work and MGD-1K baselines show that public data can support

reproducible comparison[7, 8]. Public data, however, do not automatically yield comparability. We identify three remaining problems.

(i) Patient-level leakage. The official 200-case Meiboscore test list contains 166 patients, while 326 of the remaining 800 images come from the same patients. Image-level random splits let the same patient's upper/lower eyelids and left/right eyes cross training and test. Even with a patient-level split, using the official 200 directly still leaves 116/166 test patients in the training set. Leakage is not "using more data": it lets subject identity become a memorised feature and systematically inflates generalisation.

(ii) Hidden effects of tiling. Tile height, overlap, threshold, and TTA are often buried in implementation detail or omitted. With identical weights and only tile height changed, we observe Dice moving from 0.370 to 0.655, while full-image inference drops to 0.593. The same model can therefore be narrated as strong or weak, and readers cannot tell whether the difference comes from architecture or inference configuration.

(iii) The gap between pixel metrics and clinical endpoints. Dice, HD95, and clDice describe pixels and topology, not Meiboscore or gland dropout. Six expert grading rounds let us directly compare the segmentation→R_loss→Meiboscore chain with clinical grades, and to test upper/lower eyelid differences. Prior work often excludes the lower eyelid because of incomplete exposure, which hides real clinical difficulty.

This paper therefore treats comparability and leakage audit as the primary contribution, not a novel module. Our contributions are:

(1) **A patient-level public benchmark.** On 999 deduplicated MGD-1K pairs we build a patient-independent split: 694/153/152 images (239/50/49 patients), and release the split file, the file-list signature, and the assignment signature, so that "same file list, different assignment" is no longer untraceable.

(2) **Leakage audit and quantification.** We report the patient composition of the official 200-case test list (166 patients), the 326/800 image-level leakage, and the 116/166 patient-level leakage; we construct a 60-image/50-patient clean-official subset and compare patient-level versus leaky protocols.

(3) **A five-route comparison under one protocol.** UNet, Attention U-Net, WRCANet, MGD-PriorNet, and MGD-Former are trained and evaluated with the same preprocessing, the same patient-level split, the same loss, and the same tiling and threshold. WRCANet did not converge within the 8 GB budget, so the main table ranks the four comparable routes and reports WRCANet separately as a memory/time feasibility counterexample. The four comparable routes are baselines and ablation branches, not failed attempts being replaced.

(4) **Tiling as a first-class variable and Multi-Tile Ensemble (MTE).** We report tile sensitivity from 128 to 512 and full image with identical weights, and propose MTE, which averages probabilities from 256/384/512 tiles; we report its HD95 boundary improvement (Dice did not improve) and compare it with 8× TTA in accuracy and inference cost.

(5) **Uncertainty-guided review.** We compare multi-tile variance, predictive entropy, and 8× TTA by AUROC for failure cases (Dice < 0.7) and report their limitations; variance is only weak-to-moderate and TTA8 suffers from label–signal coupling, so neither is a standalone triage rule.

(6) **Clinical grading and subgroup analysis.** We report accuracy, MAE, and quadratic weighted kappa between segmentation-derived R_loss/Meiboscore and six-round expert grades, and report Dice/clDice separately for upper and lower eyelids.

No accuracy claim is fabricated. All Dice/IoU/HD95/clDice, AUROC, and clinical agreement values will be injected by script after unified training and independent testing; tiling sensitivity and VRAM/throughput are engineering measurements labelled with their conditions. Section 3 describes the method and training recipe, Section 4 turns patient-level splitting, leakage audit, and MTE into methodological objects, Section 5 gives the experiment matrix, Section 6 presents the result structure and discussion, and Section 7 lists limitations.

2 Related Work

2.1 Clinical Diagnosis and Infrared Meibography

MGD diagnosis combines gland morphology with secretion quality[1, 2]. Infrared/near-infrared meibography exploits scattering differences to show dropout, tortuosity, shortening, and cystic dilation[3]. Meiboscore grades dropout area and is the most widely used semi-quantitative tool, but repeatability differs across infrared systems and inter-observer agreement is not guaranteed[4]. Two requirements follow: segmentation must support consistent area, length, and grading rather than visual appeal; and the eyelid margin must be segmented as well, otherwise the denominators of R_loss and Meiboscore are unstable.

2.2 Automated Segmentation of Meibomian Glands

Early work used thresholds, morphology, and strip filters, and was sensitive to low contrast and device variation. Deep learning followed: Prabhu et al. segmented glands and quantified morphology[9]; Wang et al. quantified gland morphology and linked it to clinical grading[10]; Liu et al. used scribble supervision to reduce pixel-level annotation cost[11]; Setu et al. and Wang et al. explored deeper or lighter encoder–decoder variants[6, 12]; Yu et al., Zhang et al., and Lin et al. introduced density estimation, attention, and multi-scale features[13, 14, 15]; Liang et al. discussed cross-device and cross-dataset generalisation[16].

These studies establish feasibility, but their reporting differs substantially: whole-image versus sliding-window inference, Dice-only versus morphological metrics, and often unclear train/validation/test exclusivity. We do not dispute their conclusions; we place them under one protocol, with explicit tiling, threshold, and split signature.

2.3 Public Benchmarks, Patient-Level Leakage, and Inference Confounders

MGD-1K is the most representative public meibography dataset: 1000 infrared images, gland and eyelid masks, 320 subjects, and six expert Meiboscore rounds per image[5]. A 2025 multicentre study further shows that gland segmentation can reach high grading agreement at independent centres, but explicitly notes that its model covers only the upper eyelid and a single device[7]; an MGD-1K baseline reports U-Net IoU around 0.72 and F1 around 0.82 without systematic patient-level splitting[8]. Public data are therefore available, but "public" does not imply "comparable".

Patient-level leakage is the easiest of these problems to hide at the implementation level. A subject in MGD-1K often has multiple images (upper/lower eyelid, left/right eye), and image-level random splits place different images of the same subject into training and test. Even with a patient-level split, directly using the official 200-case Meiboscore test list leaves many other images of those test patients in training. Few studies report both a file-list signature and an assignment signature, so readers cannot tell whether two experiments actually used the same split.

Tiling at inference is another hidden variable. Buglakova et al. show that stitching artefacts in sliding-window inference are tied to normalisation layers, and compare the trade-off between artifact-free and high-quality predictions[17]. For 3–8-pixel glands, tile height also changes the effective receptive field and context ratio, and its effect can exceed architecture differences. We treat tiling as a first-class variable and build a multi-tile ensemble on top of it.

For uncertainty, the decision-aware protocol of Maganti and Gowda shows that TTA often gives a stronger error-detection signal than MC Dropout under a fixed review budget, but estimator rankings are not universal[18]; Geißler et al. show that structure size confounds the relationship between uncertainty and segmentation quality[19]. In thin-structure tasks, the uncertainty source and structure size must therefore be reported together rather than as a single AUROC.

2.4 Architectures for Medical Image Segmentation

Encoder–decoder with skip connections remains a strong baseline[20]; Attention U-Net gates irrelevant skip responses[21]; UNet++ and MultiResUNet improve nested connectivity and multi-resolution processing[22, 23]; nnU-Net contributes a self-configured recipe—patch sampling, deep supervision, strong augmentation—rather than a module[24]. Among Transformers, TransUNet uses a Transformer encoder[25], Swin-UNet builds a U-shaped window-attention network[26], and SegFormer pairs a hierarchical MiT encoder with a lightweight head[27]; UNETR and UNETR++ bring Transformers to 3D medical segmentation[28, 29], and MedNeXt scales ConvNets with Transformer-inspired blocks[30]. State-space models, represented by VM-UNet and U-Mamba, model long-range dependencies with linear complexity[31, 32].

For this task, long-range dependency is not the bottleneck; detail preservation is. SegFormer's finest stage has stride 4 and offers ADE20K/ImageNet pretraining, matching the 3–8-pixel target and the 999-case regime. Swin-UNet's windowed attention and four-stage downsampling are unfriendly to thin-line continuity; Mamba variants may fragment connectivity through 2D scan order and depend on compiled CUDA kernels, raising cross-platform reproduction cost. We therefore adopt SegFormer-B2 as the encoder and modify the decoder to retain 1/4 features; the other routes serve as literature comparisons and are not the default.

2.5 Losses and Metrics for Thin Tubular Structures

Region losses (Dice, cross-entropy) are sensitive to overlap but not to connectivity: when a gland breaks into two segments, Dice may drop only slightly while HD95 rises sharply. clDice constrains topology through soft-skeleton centerline overlap[33]; skeleton-recall loss makes skeleton recall explicit[34]; boundary loss and Boundary IoU focus optimisation and evaluation

on the boundary band[35, 36]. Together they show that thin structures require multi-metric evaluation, since Dice alone hides topological failure.

We use clDice as the only skeleton-continuity term and do not add an explicit skeleton loss; the rationale is given in Section 3.4: two terms pushing the same gradient direction amplify skeleton noise. Boundary loss remains an optional switch, disabled by default.

2.6 Low-Contrast Enhancement, Domain Shift, and Inference Strategies

CLAHE is the most common local contrast enhancement for infrared and medical images; its clip limit and window size control noise amplification and artefacts[37]. Deep low-light enhancement, such as Zero-DCE, offers a data-driven alternative[38, 39], but usually targets visual quality rather than segmentation metrics. For domain shift, test-time adaptation and continual test-time adaptation adjust statistics without labels[40, 41]; TTA reduces variance by averaging geometric and scale views[42].

We do not use test-time training, to avoid introducing adaptation on the test set; TTA is fixed to horizontal flips and two scales, and is enabled only for paper evaluation and final delivery. Cross-device variation is addressed mainly by intensity augmentation and a unified normalisation contract, not by modifying the model at test time.

2.7 Interpretability, Uncertainty, and Reporting Standards

Grad-CAM and its variants map gradients back to the input[43, 44, 45]; uncertainty estimation separates aleatoric from epistemic, with TTA entropy and MC dropout as common approximations[42, 46, 47]. Reporting standards include CLAIM, TRIPOD+AI, PROBAST, and REFORMS, covering imaging AI, prediction models, bias assessment, and reproducible research[48, 49, 50, 51]; metric selection itself has systematic recommendations and pitfalls[52, 53].

We borrow their structure but operationalise the three items most relevant here: split signature, tiling-parameter reporting, and weight-status labelling. The system implements true Grad-CAM and TTA entropy maps, but we treat them as review aids rather than performance evidence.

3 Method

3.1 Task Definition and Notation

Given an infrared meibography image $I \in \mathbb{R}^{H\times W}$ with $H = 640$ and $W = 1280$, single-channel grayscale, we define two pixel-wise binary targets:

$$G \in \{0,1\}^{H\times W}, \qquad E \in \{0,1\}^{H\times W}, \tag{1}$$

where G is the gland mask and E the eyelid-margin mask. The network $f(\cdot;\theta)$ outputs two-channel logits:

$$\mathbf{z} = f(I;\theta) \in \mathbb{R}^{2\times H\times W}, \quad \hat{G} = \mathbf{1}[\sigma(z_G) \geq \tau], \quad \hat{E} = \mathbf{1}[\sigma(z_E) \geq \tau], \tag{2}$$

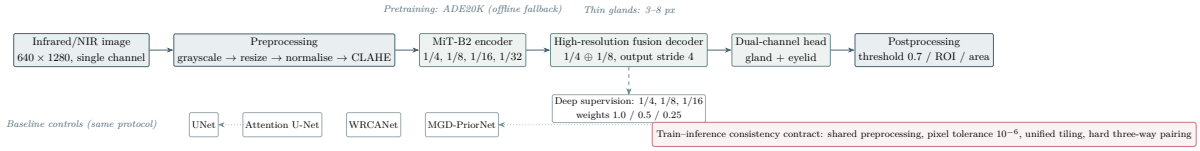


Figure 1 Method overview. The upper row is the default chain: shared preprocessing, a MiT-B2 encoder, a high-resolution fusion decoder retaining 1/4 and 1/8 features (stride 4), a dual-channel head, and postprocessing. The lower row lists four baselines compared under the same protocol. The bottom band denotes one preprocessing implementation shared by training and inference.

with σ the sigmoid and τ the threshold. Glands are the primary task; the eyelid is auxiliary, providing anatomical ROI, the denominator of R_loss, and the normalisation basis of Meiboscore. The two channels share the encoder but have independent output heads: they arise from the same imaging process with correlated low-level texture and illumination, yet their geometry differs enough that fully shared discriminative features would be harmful.

Figure 1 gives the overview. In addition to the linear acquisition–preprocessing–network–output–postprocessing chain, it shows a baseline branch: UNet, Attention U-Net, WRCANet, and MGD-PriorNet are not superseded implementations but controls evaluated under the same protocol. The bottom band denotes the contract by which training and inference share one preprocessing implementation, locked by a pixel-wise test.

3.2 Preprocessing

The goal is to map images from different devices and exposures into one numerical space while keeping training and inference identical. The pipeline has five fixed steps, whose order cannot be exchanged:

P1. Orientation. Read with PIL and apply the EXIF orientation, avoiding 90° or 180° mismatches between training and inference.

P2. Grayscale. Convert RGB/RGBA with BT.601 weights; floating input is multiplied by 255 if its maximum is ≤ 1, then clipped to $[0, 255]$ and cast to `uint8`.

P3. Resize. Bilinear interpolation to 640×1280; skip resampling if already at the target size.

P4. Intensity normalisation. Divide `uint8` by 255 to obtain $[0, 1]$ floats.

P5. Local contrast enhancement. CLAHE with clip limit 0.02, producing $[0, 1]$ `float32`.

Labels follow a separate path: same orientation and resize, but nearest-neighbour interpolation, then binarisation at > 0. Nearest-neighbour prevents thin glands from becoming grey edges; binarisation ensures the network only sees 0/1 supervision. The path applies to both gland and eyelid labels. Figure 2 contrasts the two paths and marks the consistency contract.

CLAHE uses scikit-image's `equalize_adapthist`, with a local window of about 1/8 of the image and 256 histogram bins. Because libraries define the clip limit differently (normalised frequency versus pixel count), the weight manifest records the library name, parameter value, and the normalisation tag `scale01+clahe`, rather than a bare number.

3.3 MGD-Former: Encoder and High-Resolution Fusion Decoder

The bottleneck here is not long-range dependency but **detail preservation** and **cold start on small data**. At 640×1280, glands are 3–8 pixels wide. If the finest scale were stride 16, the

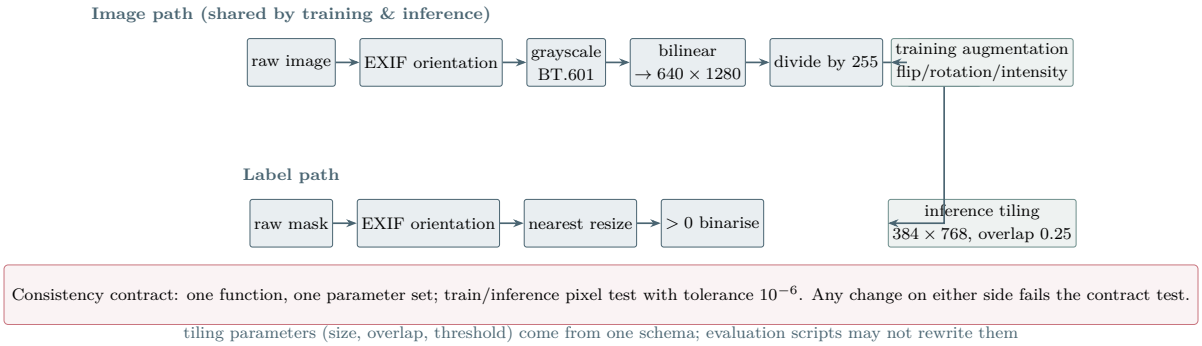


Figure 2 Preprocessing consistency contract. The upper lane is the image path; the lower lane is the label path (nearest-neighbour + binarisation). The right side shows training augmentation and inference tiling. Training and inference call one implementation, locked by a pixel-wise test; tiling and threshold come from one schema.

feature map would be only 40×80, and a 4-pixel gland would occupy less than 0.25 pixels—it would disappear during downsampling. HD95 is highly sensitive to breaks, and such structural loss cannot be recovered by postprocessing. We therefore adopt an architecture with **output stride 4** that feeds the 1/4 feature directly into the decoder.

MGD-Former has two parts. The **encoder** is MiT-B2[27]: four stages with strides 4/8/16/32, dimensions 64/128/320/512, depths 3/4/6/3, attention heads 1/2/5/8, and spatial-reduction ratios 8/4/2/1. Stage 1 uses overlapping 7×7 convolutions with stride 4; stages 2–4 use overlapping 3×3 convolutions with stride 2. Each block combines efficient self-attention with a Mix-FFN that inserts a 3×3 depthwise convolution between two linear layers, injecting positional information without fixed positional encodings that break under resolution change. DropPath increases linearly from 0 to 0.1.

The **decoder** does not use SegFormer's lightweight MLP head. It builds a top-down fusion structure: all four stages are projected to 128 channels by 1×1 convolutions; starting from 1/32, each level is upsampled, added to the next finer feature, and passed through a fusion block (3×3 convolution + BatchNorm + GELU), yielding fused features at 1/16, 1/8, and 1/4. A final 3×3 convolution, BatchNorm, GELU, and 1×1 convolution produce two-channel logits, which are bilinearly upsampled to 640×1280. Because the last fusion is at 1/4, the effective output stride is 4: a thin gland still occupies roughly 1–2 feature cells at decoding time instead of collapsing below a pixel.

Deep supervision is attached to the 1/4, 1/8, and 1/16 fused features, each with a 1×1 auxiliary head supervising only the gland channel; auxiliary losses are weighted 1.0/0.5/0.25 and normalised. This stabilises gradients on small data and forces intermediate scales to localise glands rather than leaving all discrimination to the final layer. Figure 3(a) shows the structure.

For **pretraining**, we use ADE20K semantic-segmentation weights (SegFormer-B2; encoder only, weight identifier in the appendix), with ImageNet as an alternative. ADE20K is preferred to training from scratch because 999 paired cases cannot teach generic edge, texture, and shape priors to a 20M-parameter Transformer; ADE20K already provides low-level filters sensitive to thin structures and local contrast, and fine-tuning only has to learn the semantics of "gland". If the target machine is offline and the local cache lacks the weights, the encoder degrades to

random initialisation; the training log and run metadata record `pretrained_status` and the reason. Reports must state whether pretraining was actually loaded and its coverage.

3.4 Loss Function and Deep Supervision

The objective combines region overlap, pixel classification, skeleton continuity, and an auxiliary task:

$$\mathcal{L}_g = 0.4\,\mathcal{L}_{\text{Dice}}(P_g, Y_g) + 0.4\,\mathcal{L}_{\text{CE}}(P_g, Y_g) + \lambda_{cl}(t)\,\mathcal{L}_{\text{clDice}}(P_g, Y_g), \tag{3}$$

$$\mathcal{L}_e = 0.4\,\mathcal{L}_{\text{Dice}}(P_e, Y_e) + 0.4\,\mathcal{L}_{\text{CE}}(P_e, Y_e), \tag{4}$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_g + 0.25\,\mathcal{L}_e + \frac{1}{1.75}\sum_{s\in\{4,8,16\}} w_s\,\mathcal{L}_s, \qquad (w_4, w_8, w_{16}) = (1.0, 0.5, 0.25), \tag{5}$$

where P_g, P_e are sigmoid probabilities, Y_g, Y_e the labels, and $\mathcal{L}_s$ the Dice+CE of the auxiliary head at scale s. The clDice weight rises linearly from 0 to 0.2 over the first 10% of epochs:

$$\lambda_{cl}(t) = 0.2 \cdot \min\left(1, \frac{t}{0.1\,T}\right), \tag{6}$$

with T the total number of epochs. Warmup prevents skeleton noise from dominating early gradients: when predictions are fragmented, differentiable-skeleton topology is unreliable, and a large weight would push the network toward a "merge first, refine later" solution.

Why no separate skeleton-consistency term. clDice is already a differentiable skeleton-consistency constraint: iterative soft erosion and opening extract centerlines from prediction and reference, and centerline overlap forms the loss[33]. Adding an explicit skeleton loss would double-weight the same gradient direction and turn skeleton-extraction noise—especially endpoint jitter on 3–8-pixel glands—into a dominant gradient source. We therefore assign skeleton continuity to exactly one term, and present this as an active design trade-off: *rather have one fewer hyperparameter than two near-redundant terms amplifying each other's noise.* Boundary loss remains an optional switch (default weight 0) for HD95-sensitive scenarios; the main configuration disables it to avoid duplicating clDice's boundary response.

The eyelid auxiliary weight 0.25 is not decorative. It yields three concrete benefits: the eyelid channel provides anatomical ROI and suppresses false positives outside the lid; R_loss uses eyelid area as its denominator, so a stable margin mask is required; and Meiboscore normalisation needs the margin extent. The ablation closes this branch to test the gain.

3.5 Baseline Architecture Family

All four baselines share the same input–output contract (single-channel input, two-channel logits, no internal sigmoid); only the backbone and skip handling differ. They provide controls under one protocol, not failed alternatives.

- **U-Net**[20]: five-level encoder–decoder, channels $(64, 128, 256, 512, 1024)$, two residual units per level, PReLU, InstanceNorm, dropout 0.1. It represents the classical "convolution + skip" route.
- **Attention U-Net**[21]: attention gates on skips, $\alpha = \sigma(\psi(\text{ReLU}(W_g * g + W_x * x)))$, $x' = \alpha \odot x$, using coarse context to suppress irrelevant regions. It represent spatial re-weighting.

- **WRCANet**: an enhancement backbone built from residual channel attention blocks (RCAB)[54]. The backbone lifts the input to 64 channels, stacks 8 residual groups of 10 RCABs, returns to one channel with a 3×3 convolution, and adds a global residual; RCAB is

$$u = \mathrm{ReLU}(\mathrm{Conv}_{3\times3}(x)), \quad v = \mathrm{Conv}_{3\times3}(u), \tag{7}$$

$$s = \sigma(W_2\,\mathrm{ReLU}(W_1\,\mathrm{GAP}(v))), \quad y = s \odot v + x, \tag{8}$$

  with reduction ratio $r = 16$, followed by a three-level U-shaped head. It represents "enhancement backbone + lightweight head".
- **MGD-PriorNet**: this project's structural-prior branch for thin glands. The FCAS front end enhances low-contrast strips with a Laplacian high-frequency residual and channel attention; ASAB runs $1 \times k$ horizontal, $k \times 1$ vertical, and 3×3 convolutions in parallel to model anisotropy; APG-DD uses multi-scale eyelid probabilities as a soft gate $(1 + A_{\mathrm{eyelid}})$ on the gland decoder. It represents "anatomical priors encoded in the network".

Positioning matters: MGD-Former contributes an adaptation of a pretrained Transformer's high-resolution stage to this task, not a claim that the four baselines are worse in absolute terms. They differ in parameters, inference cost, and data requirements; Section 6 provides comparable evidence on one test set. Figure 3(b) summarises the four families and details RCAB.

3.6 Training Recipe and Inference Strategy

The recipe is constrained by small data, cross-device variation, and consumer GPUs:

- **Precision and memory**: AMP on CUDA; disabled on CPU and logged. channels_last is on by default; gradient accumulation emulates a larger batch on 8 GB cards.
- **EMA**: decay 0.999; evaluation and saving use EMA weights. At 999 cases, online weights have high variance, and EMA is a low-cost stabiliser.
- **Optimiser**: AdamW, learning rate 1×10^{-4} for the pretrained encoder, weight decay 1×10^{-4}; linear warmup over 5% of steps, then cosine decay to 1% of the initial rate (10^{-6}). Randomly initialised baselines may use 10^{-3}.
- **Augmentation**: elastic deformation, rotation $\pm10°$, scale 0.85–1.20, **horizontal flip only**, intensity perturbation (gamma 0.8–1.25, CLAHE clip 0.01–0.03, Gaussian noise, 0–1 pixel blur). Intensity perturbation targets cross-device shift and matters more here than geometric augmentation.
- **Why vertical flip is disabled**: glands run near-vertically, and upper and lower lid anatomy is asymmetric. A vertical flip turns "upper-lid glands" into anatomically impossible "lower-lid glands" and breaks the relative-position prior between margin and glands. A horizontal flip preserves the vertical orientation and the upper/lower relationship, and is therefore safe and effective.
- **Multi-scale training**: random 0.75/1.0/1.25 scaling while preserving the 2:1 aspect ratio, so the model sees different gland widths and depends less on one resolution.
- **TTA**: three levels—off, 2× (horizontal flip), and 8× (four flips × two scales: native and 1.125×). The 8× setting is used only for paper metrics and final delivery, not for training-time monitoring, so that validation curves are not bought with inference budget.

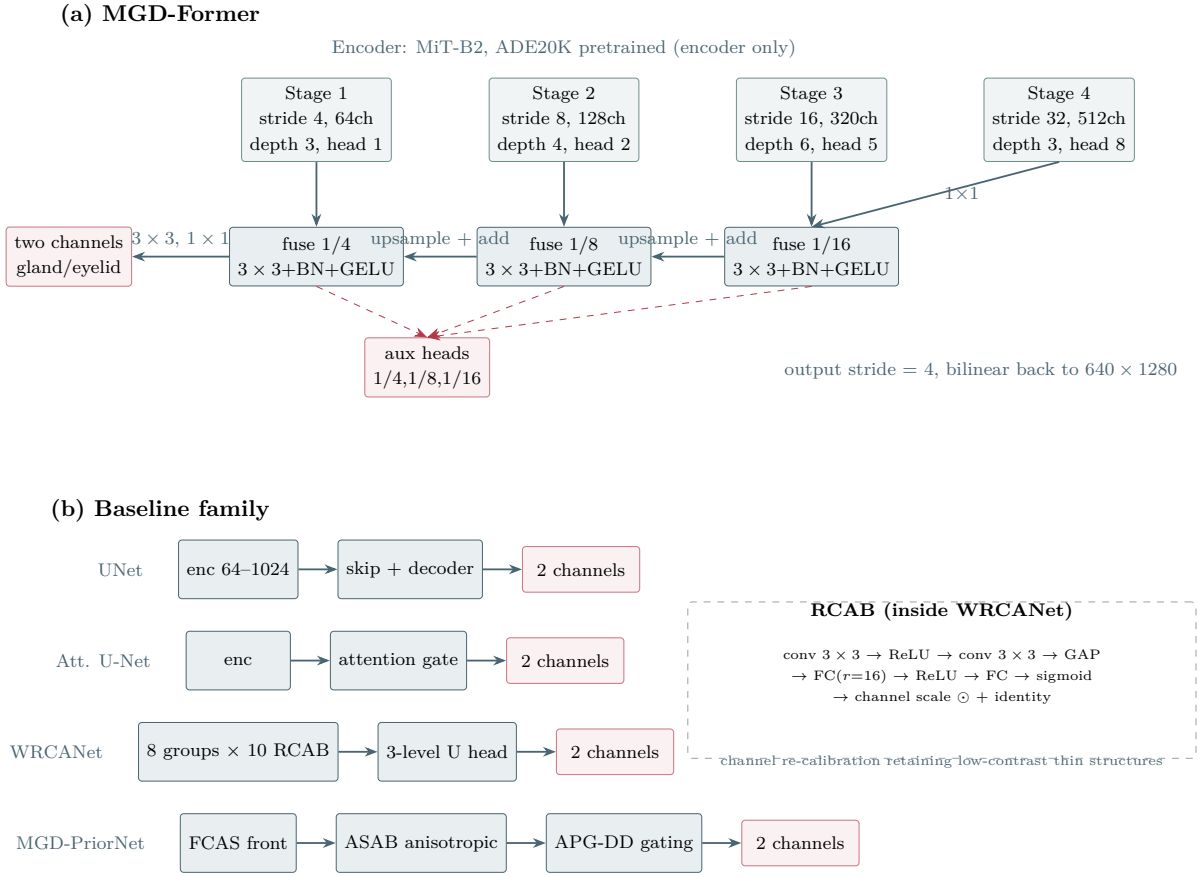


Figure 3 Network structures. (a) MGD-Former: a four-stage MiT-B2 encoder; the 1/32 feature is fused successively with 1/16, 1/8, and 1/4 features, producing stride 4; three auxiliary heads attach at 1/4, 1/8, and 1/16. (b) The four baselines: symmetric UNet skips, Attention U-Net gating, WRCANet's RCAB enhancement backbone with a three-level U head, and MGD-PriorNet's FCAS/ASAB/APG-DD.

- **Ensemble**: probability averaging of five fold weights as the final model; single-fold and ensemble results are reported separately.

Inference defaults match training: tiles of 384×768, overlap 0.25, threshold 0.7. ROI constraint, a 120-pixel minimum component, and Gaussian smoothing $\sigma = 1.2$ are postprocessing; whether they are enabled must be stated in results, and Section 6 separates "network capability" from "postprocessing gain".

3.7 Evaluation Metrics and Statistical Protocol

With prediction P and reference Y, Dice and IoU are

$$\text{Dice} = \frac{2|P \cap Y|}{|P| + |Y|}, \qquad \text{IoU} = \frac{|P \cap Y|}{|P \cup Y|}. \tag{9}$$

Surface distances use HD95 (95th percentile of the bidirectional surface distance) and ASSD (bidirectional mean). Skeleton continuity uses clDice, sharing its origin with the differentiable training term but computed on discrete skeletons at evaluation. We also report Precision, Recall, eyelid Dice, R_loss, and Meiboscore agreement.

Three caveats. First, images lack physical calibration, so HD95 and ASSD are in **pixels**;

after resizing to 640 × 1280, values are not directly comparable across devices or original resolutions. Second, surface distance is undefined when either mask is empty; we return NaN and record the reason rather than reporting 0 as perfect. Third, aggregation ignores NaN and reports the number of valid cases n.

Means are reported with 95% confidence intervals from 1,000 bootstrap resamples. Model comparisons use the per-sample paired Wilcoxon signed-rank test with Holm correction. The significance level is 0.05, but p values are not the only evidence: confidence intervals and effect direction matter equally. Every report must carry `split_signature`; a missing signature invalidates the report (Section 4.7).

3.8 Evaluation Scope: Train-Self versus Independent Test

We define two scopes explicitly:

- **Train-self (`train_self`)**: evaluation on samples used for training. It answers only whether the model fits the training data and does not represent generalisation; such values appear in a separate column/table labelled "not generalisation".
- **Independent test (`holdout`)**: evaluation on samples never used for training, early stopping, or model selection. Only this scope supports performance conclusions.

This distinction is not formalism. The legacy UNet weights were trained after shuffling all 999 cases with no held-out test set, so any "test" metric may be train-self. Section 6 places it only in the same-distribution column and states this explicitly.

3.9 Patient-Level Split and Leakage Audit

MGD-1K filenames contain a numeric study id (e.g. `..._11756049`); upper/lower eyelids and left/right eyes of the same subject share it. From 999 deduplicated pairs we extract 338 ids, group by id, stratify each group by its maximum expert Meiboscore, and assign whole subjects to train/val/test, obtaining 694/153/152 images (239/50/49 patients) with mutually exclusive patients. The split file records two digests: the file-list signature 6bdacfb0a71b8196 08dba4cde4382f8dd23fb4a88b72c9b6d151a99ed83bc22a (identifying the 999 stems only) and the assignment signature 7a6fd5819d048e0ccd811c943d616a0fe8738a5c06bc78752c66bb38624795f6 (SHA-256 of the ordered train/val/test lists). The former answers "did the dataset change?"; the latter answers "which assignment did this run actually use?". Without the latter, the same file list can be re-assigned without traceability.

The leakage audit has three levels. First, the official 200-case Meiboscore test list contains 166 patients; 326 of the remaining 800 images come from those patients. Second, even with a patient-level split, the 694 training set still contains 116/166 official-test patients (355 images). Third, we build a clean-official subset containing only official-test images whose patient is absent from the patient-level training set: 60 images/50 patients. We additionally train an image-level leaky model (649 train / 150 val, official 200 test) and compare it with the patient-level model on the same clean-official subset to quantify the leakage gap.

3.10 Multi-Tile Ensemble and Uncertainty

For input I, tile height $h \in \{256, 384, 512\}$, width $2h$, overlap 0.25, and threshold $\tau = 0.7$, sliding-window inference produces P_h. MTE outputs

$$\bar{P} = \frac{1}{3} \sum_{h \in \{256,384,512\}} P_h, \tag{10}$$

which is then thresholded at τ. MTE does not change the weights or add training cost; inference costs about three single-tile passes, less than 8× TTA. We report Dice/HD95/clDice relative to the default 384 tile and compare with 8× TTA.

For uncertainty, the three P_h of one image give a pixel variance $V = \mathrm{Var}_h(P_h)$ and the Shannon entropy $H(\bar{P})$ of the mean probability. Defining failure as Dice < 0.7, we compute the AUROC of V, H, and the 8× TTA error $(1 - \mathrm{Dice}_{\mathrm{TTA8}})$, using a tie-aware rank sum. This treats "changing the tile" itself as a low-cost uncertainty source, to be compared with TTA.

3.11 Clinical Grading Endpoints

Given a predicted gland mask M_g and the reference eyelid mask M_e,

$$R_{\mathrm{loss}} = \left(1 - \frac{|M_g \cap M_e|}{|M_e|}\right) \times 100\%, \tag{11}$$

and Meiboscore is mapped by the 0/1/2/3 thresholds. We compare with the majority vote of the six expert Meiboscore rounds and report accuracy, MAE, and quadratic weighted kappa (QWK). We also report Dice/clDice separately for upper and lower eyelids to test whether the lower eyelid is significantly harder.

4 Engineering as Method: Consistency, Tiling, and Reproducibility

This section is not a system-implementation appendix. We argue that in thin-structure segmentation, several variables treated as engineering detail materially change conclusions; elevating them to methodological objects and fixing them with contracts and protocols is itself part of reproducible research. The five subsections address consistency, tiling, memory, pipelines, and evaluation discipline, followed by the layered architecture.

4.1 Patient-Level Split and Leakage Audit

A subject in MGD-1K often has multiple images (upper/lower eyelid, left/right eye), and the numeric study id in the filename provides an executable patient identifier. We assign whole subjects, not individual images, to train/val/test, obtaining 694/153/152 images (239/50/49 patients) with mutually exclusive patients; stratification uses each subject's maximum expert Meiboscore so that severe cases are not missing from one split. The split file records both the file-list signature 6bdacfb0a71b819608dba4cde4382f8dd23fb4a88b72c9b6d151a99ed83bc22a and the assignment signature 7a6fd5819d048e0ccd811c943d616a0fe8738a5c06bc78752c66bb38624795f6: the former answers whether the dataset changed, the latter answers which assignment this run

used. Without the latter, the same file list can be re-assigned without traceability, which is the root of "same names, different split".

The leakage audit provides three levels of evidence. First, the official 200-case Meiboscore test list comes from 166 patients; 326 of the remaining 800 images come from those patients. Second, even with a patient-level split, the 694 training set still contains 116/166 official-test patients (355 images). Third, we build a clean-official subset containing only official-test images whose patient is absent from training: 60 images/50 patients, and train an image-level leaky control (649 train / 150 val). The difference among the three on the same clean-official subset is the cost of leakage. Figure 4 relates pairing, patient grouping, and the two signatures.

4.2 Train–Inference Consistency Contract

Performance differences in deep segmentation are often attributed to networks, yet the same weights can yield very different Dice under different preprocessing, tiling, or thresholding. To remove this hidden variance, we turn "what training sees" and "what inference sees" into an executable contract rather than documentation.

The contract has three hard rules.

C1. Pixel-wise preprocessing identity. Online training transforms and inference preprocessing call the same implementation: grayscale reading, EXIF orientation, bilinear resize to 640×1280, division by 255, and CLAHE (clip 0.02) are shared functions; training only appends random augmentation after the shared function. A unit test compares training and inference paths on fixed inputs pixel by pixel with tolerance 10^{-6}; a change on either side fails the test.

C2. Unified tiling parameters. Training tiles, evaluation tiles, and deployment defaults come from one schema (default 384×768, overlap 0.25). Evaluation scripts may not rewrite tiling; an experiment that changes it must record the change and be treated as a separate experiment (Section 4.3).

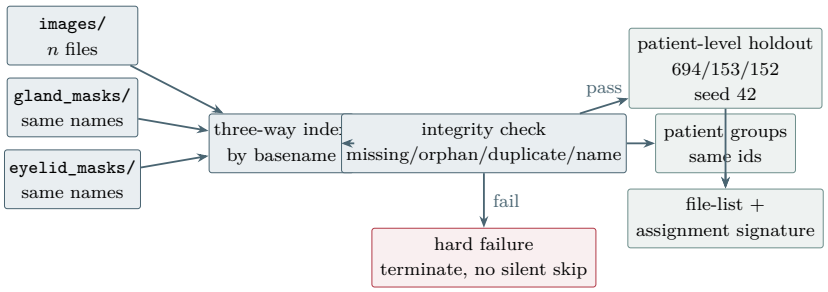
C3. Hard failure on three-way pairing. Images, gland masks, and eyelid masks are indexed by basename. A missing item, orphan mask, duplicate name, or malformed name raises an integrity error and terminates loading instead of skipping samples. Silent skipping would make the effective sample set differ across experiments and destroy comparability.

Figures 2 and 4 show the contract and the pairing/splitting flow. The benefit is not cleaner code but the guarantee that all model comparisons in Section 6 share one input distribution; any run violating the contract leaves inspectable traces in its artefacts.

4.3 The Hidden Effect of Tile Size

Experimental protocol. All numbers in this subsection come from inference with **the same weights and only a changed tile height**: threshold 0.7, TTA off, no postprocessing (no ROI constraint, no connected-component filtering, no morphological smoothing), evaluated on 40 original images. Apart from tile height, preprocessing, weights, threshold, and evaluation code are identical. This protocol is a precondition for the conclusion; any change in these factors must not be compared directly with the table below.

Table 1 and Figure 5 give seven measured points. Three observations matter. First, Dice



The signature is written into every evaluation report; a report without one is invalid and may not enter paper tables or weight promotion

Figure 4 Three-way pairing, integrity, and patient-level splitting. Images, gland masks, and eyelid masks are indexed by basename; an integrity failure terminates the run. The patient-level persistent holdout uses 694/153/152 (239/50/49 patients) and records both a file-list signature and an assignment signature; five-fold is an extension reusing the same patient groups.

Table 1 Effect of tile height on Dice (same weights, only tiling changed; threshold 0.7, TTA off, no postprocessing or ROI; $n = 40$).

Tile height (pixels)	128	192	256	384	448	512	full 640 × 1280
Dice	0.370	0.410	0.480	**0.644**	0.643	0.655	0.593

rises by about 27 percentage points from 128 to 384—not a network difference but an inference-configuration difference with identical weights. Second, 512 is slightly above 384, yet full-image inference drops to 0.593: the model was trained on 384×768 tiles, and full-image input changes the effective receptive field and context ratio, so train–inference scale no longer matches. Third, the plateau around 384 (0.644/0.643/0.655) shows saturation; increasing tiles further is not monotonically better.

The default was therefore changed from earlier 192/256 to 384×768, and every report must record tile height and overlap. The methodological point is that a paper reporting only its "best configuration" can make the same model look strong or weak, and readers cannot reproduce it; tiling must be treated as an experimental variable on par with backbone and loss.

4.4 Multi-Tile Ensemble: Turning Tiling from a Confounder into an Inference Tool

Tile sensitivity means that conclusions from a single tile do not transfer. Rather than reporting only one optimal tile, we average probability maps from multiple tiles: for $h \in \{256, 384, 512\}$, sliding-window inference gives P_h, and MTE outputs $\bar{P} = \frac{1}{3}\sum_h P_h$, thresholded at 0.7. MTE does not change the weights or add training cost; it costs about three single-tile passes, less than 8× TTA. Its purpose is not "a little more Dice" but to turn tiling from an uncontrolled confounder into an explicit inference tool: when 256 and 512 disagree, MTE gives a stable estimate that does not depend on a single choice.

The pixel variance of the three P_h, $V = \mathrm{Var}_h(P_h)$, is a low-cost uncertainty source. We use it together with predictive entropy and 8× TTA error for failure detection (Dice < 0.7) and report AUROC. Tiling is thus both the object of audit and the tool for auditing other variables.

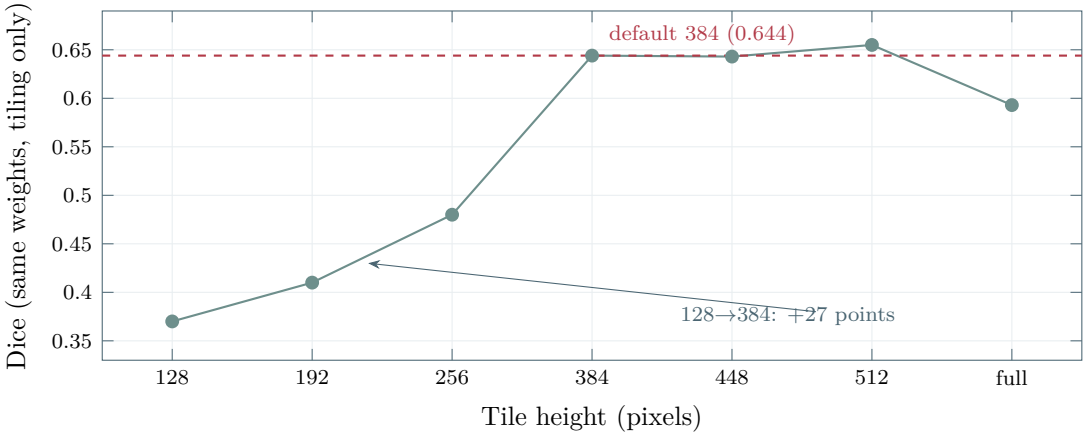


Figure 5 Tiling sensitivity (same weights, only tile height changed; threshold 0.7, TTA off, no postprocessing or ROI; $n = 40$). The default is 384; full-image inference drops because the train–inference scale no longer matches.

Table 2 VRAM tier recipes. The 6 GB and 8 GB tiers are measured on the pre-deduplication 700-case training set; 12 GB and above are extrapolated estimates and must not be cited as measurements.

Tier	Tile	batch/accum	Deep sup.	Peak VRAM	Throughput (samples/s)	Status
6 GB	256 × 512	1/4	No	0.89 GB	3.54	measured
8 GB	384 × 768	1/4	Yes	1.227 GB	3.10	measured
12 GB	384 × 768	2/2	Yes	estimate	estimate	estimated
16 GB	512 × 1024	2/1	Yes	estimate	estimate	estimated
24 GB	640 × 1280	2/1	Yes	estimate	estimate	estimated
40 GB+	640 × 1280	4/1	Yes	estimate	estimate	estimated

4.5 VRAM Tiers and Consumer-GPU Reproducibility

To make "reproducible on a consumer GPU" testable rather than rhetorical, we provide six memory tiers (Table 2, Figure 6). Each tier fixes tiling, batch, gradient accumulation, AMP, channels_last, deep supervision, and TTA, and is labelled **measured** or **estimated**. Measurements come from an RTX 4060 Laptop 8 GB, Python 3.11, torch 2.5.1 CUDA 12.1, with 700 training samples (pre-deduplication). The current deduplicated split has 699 training cases; a one-case difference does not affect the throughput conclusion.

The 8 GB tier measured 225.6 s/epoch, 3.10 samples/s, and 1.227 GB peak (run `20261106T203908Z_MGD-Former_nogit_7c6ef4`); the 6 GB tier measured 197.5 s/epoch, 3.54 samples/s, and 0.89 GB peak. All epoch times, throughputs, and VRAM peaks above were measured on the same 700-case pre-deduplication training set. Both smoke runs used random initialisation because pretrained weights were unavailable offline, and the log records this degradation. These numbers answer "can it run on 8 GB and at what throughput", not "what accuracy after pretraining".

A tier is a safety boundary, not a suggestion: selecting it overrides tiling, batch, and AMP fields, preventing "starts but must OOM" combinations. After moving machines, the first run must inspect `peak_vram_gb`; measured values are then back-filled and the status changed. Estimated tiers must never be presented as measured. This rule shares its rationale with

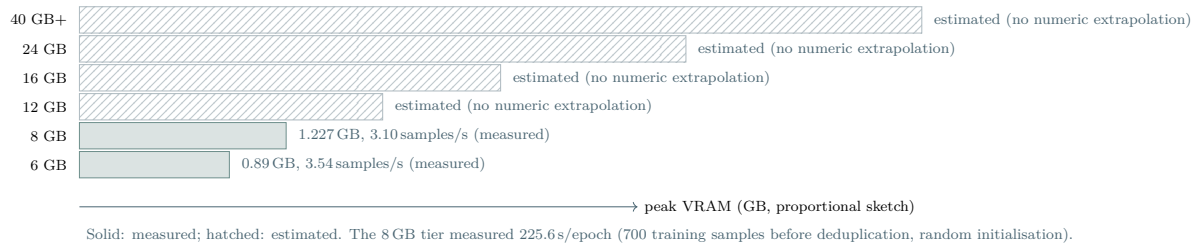


Figure 6 Six VRAM tiers. The 6 GB and 8 GB tiers are measured on an RTX 4060 Laptop 8 GB with the 700-case pre-deduplication training set (peak and throughput); 12 GB and above are planning tiers with no numeric extrapolation. Estimated tiers must not be presented as measurements.

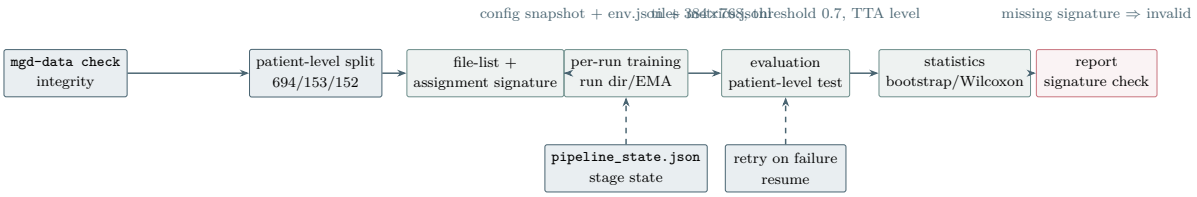


Figure 7 Experiment pipeline. Integrity checking and splitting come first, and the signature is written into each run. Per-fold training produces an independent run directory; evaluation and statistics use out-of-fold predictions; the report stage validates the signature and refuses reports without one. The state file supports retry and resumption.

Section 4.7: reproducibility depends on explicitly separating the known from the extrapolated.

4.6 Orchestrated Pipeline and Experiment Registry

Model comparison requires multi-fold, multi-configuration, interruptible jobs, and manual notes easily lose context. We organise training, evaluation, statistics, and reporting into a pipeline with stages `pretrain/train/finetune/eval/ablation/report`; state is written to `pipeline_state.json`, and a resumed run skips completed entries while retrying failures according to configuration.

Each run gets an independent directory named

$$\texttt{run_id} = \text{timestamp_name_code version_config fingerprint (6 chars)}, \tag{12}$$

with automatic suffixes on collision and no overwriting. The directory contains a configuration snapshot `config.yaml` (fully resolved), an environment snapshot `env.json` (code version, git commit, dependencies, device, paths), per-epoch `metrics.jsonl`, full-state checkpoints, a `reference_report.json` with `split_signature`, and `run_result.json` (peak VRAM, throughput, wall time). Resumption saves model, optimiser, scheduler, AMP scaler, epoch, global step, RNG state, EMA shadow, and data signature.

The registry makes multi-configuration comparison machine-readable: the same `metrics.jsonl` and `run_result.json` support both the paper tables and the traceability of failed runs. The experiment matrix in Section 5 maps directly onto these fields rather than relying on handwritten notes.

4.7 Evaluation Discipline and `split_signature`

We write evaluation as a protocol, not "run the test set once". It has five rules:

E1. Patient-level persistent splits. Holdout uses 694/153/152 (239/50/49 patients, seed 42), written to `dataset_splits.json` with both a file-list signature and an assignment signature; five-fold cross-validation reuses the same patient groups and does not re-split.

E2. Mandatory signature. `split_signature` is the SHA-256 digest of the ordered train/validation/test filename sets. Every evaluation report must carry it; a report without one is invalid and may not enter paper tables or weight promotion.

E3. Scope separation. `train_self` must be labelled and can only describe fit; only holdout or out-of-fold predictions support performance conclusions.

E4. Statistical completeness. Means carry 95% confidence intervals from 1,000 bootstrap resamples; models are compared with per-sample paired Wilcoxon signed-rank tests and Holm correction; the number of valid cases is reported.

E5. Weight status. Training artefacts are marked `verified=false`; only weights with a valid signature and independent evaluation can be promoted. Loading unverified weights requires explicit permission and a stated reason in logs and reports.

The signature rule guards against three real risks: silently re-splitting after tuning, so that the test set effectively participates in model selection; comparing experiments whose effective sample sets differ because corrupted samples were skipped; and presenting legacy weights trained on all data as independent test results. The signature turns "which samples participated in what" into immutable metadata, excluding these risks by construction.

4.8 Layered Architecture and Deployment

The system separates kernel and adapters: `core` has no UI dependency and contains configuration, runtime, data, models, training, inference, evaluation, imaging, reporting, and workflow; `adapters` provide thin CLI and WebUI shells over the same API. Two delivery targets (Windows WebUI and Linux CLI) share one kernel at version 2.3.0, with 146 contract and regression tests passing.

A model registry manages MGD-Former and the four baselines; `unwrap_outputs()` unifies the output contract (legacy models return a tensor, MGD-Former returns `logits` plus three auxiliary heads). The weight manifest records filename, architecture, version, SHA-256, input size, normalisation tag (`scale01+clahe` with clip limit), training provenance, and `verified` status. Device adaptation covers CUDA/CPU; the inference service caches a model singleton and supports explicit release. Figure 8 shows the layers and data flow.

These details belong in the method section because they determine whether others can rerun the experiments: one manifest, one configuration snapshot, one set of tiling defaults, and one signature form the traceable result–code–data chain. Without any link, the comparisons in Section 6 are only unverifiable numbers.

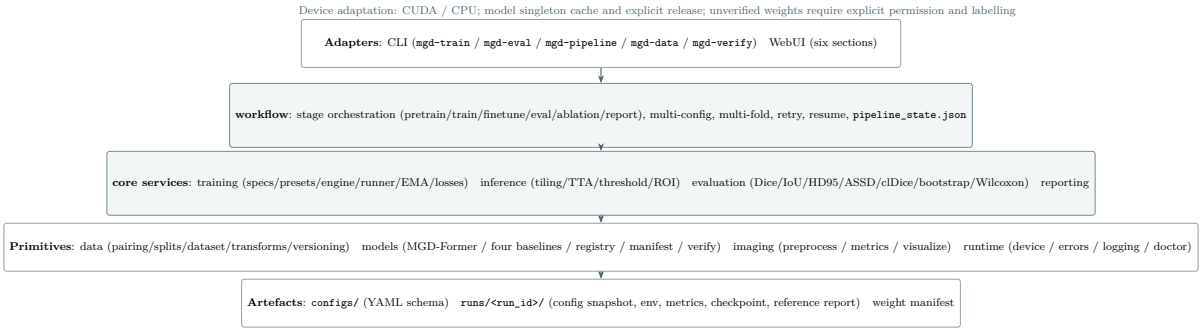


Figure 8 System layers and data flow. Adapters are thin shells; every entry point passes through the same training, inference, and evaluation implementation. Workflow handles orchestration and resumption; the artefact layer stores configuration snapshots, per-epoch metrics, full checkpoints, and signed evaluation reports.

5 Experimental Design

This section specifies an executable experiment matrix. All experiments share the consistency contract, tiling defaults, and evaluation protocol of Section 4; the only varying factor is stated in each table. Result tables are populated from the evaluation output by script; positions without a measurement remain "—", with a note on which artefacts are required.

5.1 Data and Cohort

The public MGD-1K dataset provides 1000 LipiView II infrared meibography images, gland and eyelid masks, and demographics for 320 subjects[5]. Of these, 467 are upper eyelids and 533 lower eyelids; 941 are gradable across six expert rounds, and every image has six Meiboscore rounds. Original sizes vary and are resized to 640×1280; images are single-channel grayscale.

Integrity is enforced by three-way pairing: missing images, missing gland masks, missing eyelid masks, orphan masks, duplicate names, and malformed names all terminate the pipeline. After pairing we apply **content-level deduplication**: SHA-256 is computed for the image and both masks of every case and compared across the corpus. One group of differently named cases shares an image and both masks byte-for-byte; removing the redundant copy yields 999 valid pairs. Source data are unchanged, and the deduplication record is written to `dataset_splits.json` and `dataset_version.json`.

5.2 Patient-Level Split and Leakage Protocol

The numeric study id in MGD-1K filenames is a patient identifier. We group the 999 cases into 338 ids, stratify each group by its maximum expert Meiboscore, and assign whole subjects to train/val/test, obtaining 694/153/152 images (239/50/49 patients) with mutually exclusive patients; the Meiboscore distribution is given in Table 3. The split file records both the file-list signature 6bdacfb0a71b819608dba4cde4382f8dd23fb4a88b72c9b6d151a99ed83bc22a and the assignment signature 7a6fd5819d048e0ccd811c943d616a0fe8738a5c06bc78752c66bb38624795f6; the former identifies the 999 stems, while the latter hashes the ordered train/val/test lists and answers which assignment this run actually used.

The leakage protocol has three levels:

Table 3 Patient-level split and expert Meiboscore distribution.

Split	Images	Patients	Meiboscore 0/1/2/3
Train	694	239	119/432/117/26
Val	153	50	26/92/28/7
Test	152	49	35/87/23/7

File-list signature: `6bdacfb0a71b819608dba4cde4382f8dd23fb4a88b72c9b6d151a99ed83bc22a`; assignment signature: `7a6fd5819d048e0ccd811c943d616a0fe8738a5c06bc78752c66bb38624795f6`.

Table 4 Experiment matrix for the 9-hour budget. All runs share the patient-level split, preprocessing, loss, and inference protocol; the leaky control uses an image-level split.

ID	Model	Split	Epochs	Effective batch	Purpose
E1	MGD-Former (ADE20K)	patient-level	45	4	main model
E2	UNet	patient-level	30	4	classical baseline
E3	AttentionUNet	patient-level	24	4	attention baseline
E4	WRCANet	patient-level	12	4 (2×2)	enhancement baseline
E5	MGD-PriorNet	patient-level	24	4	anatomical-prior baseline
E6	UNet w/o clDice	patient-level	—	4	not run (compute moved to AttentionUNet continuation)
E7	UNet	official-200 leaky	25	4	leakage control

Actual epochs are taken from `metrics.jsonl`: MGD-Former 45, UNet 30, AttentionUNet 15 (9 + 6 continuation), WRCANet 1 (budget-truncated, not ranked), MGD-PriorNet about 12 (budget-truncated). The table lists planned maxima.

L1. Patient composition of the official test list. The official 200-case Meiboscore test list comes from 166 patients; 326 of the remaining 800 images come from the same patients.

L2. Residual leakage in the patient-level training set. Even with a patient-level split, the 694 training set still contains 116/166 official-test patients (355 images).

L3. Clean-official control. We keep only official-test images whose patient is absent from the patient-level training set: 60 images/50 patients. We additionally train an image-level leaky control (649 train / 150 val, official 200 test) and compare it on the same clean-official subset.

Threshold 0.7, tiles 384×768, and overlap 0.25 are fixed once on validation and frozen; no test-time tuning is permitted. If a model requires a different threshold, it is reported as part of that model's configuration.

5.3 Compared Models and Training Budget

The main table includes five models: UNet, Attention U-Net, WRCANet, MGD-PriorNet, and MGD-Former. All use the same loss (Section 3.4), augmentation, optimiser, and patient-level split; only backbone and decoder differ. To keep the 9-hour budget, this round uses a single holdout rather than five-fold cross-validation; five-fold is left as an extension and is not presented as if it had been run. Table 4 lists the seven training runs actually executed.

Recipe: AdamW, learning rate 1×10^{-4}, weight decay 1×10^{-4}, cosine annealing, AMP and channels_last on, EMA 0.99. EMA is 0.99 rather than 0.999 because under a 9-hour budget the 0.999 shadow is still dominated by random initialisation after a few hundred steps, hiding the validation curve behind the shadow weights; this is logged and is not the long-training default. The WRCANet enhancement backbone uses gradient checkpointing on the 8 GB card with batch 2 and accumulation 2; other models use batch 4 and accumulation 1, giving an

effective batch of 4.

5.4 Metrics and Statistics

For glands we report Dice, IoU, HD95, ASSD, clDice, Precision, and Recall; for the eyelid auxiliary task we report Dice. Surface distances are in pixels because images lack physical calibration. Undefined values from empty masks are returned as NaN with a reason, never replaced by 0. Clinical endpoints include R_loss, Meiboscore, and accuracy/MAE/quadratic weighted kappa (QWK) against six expert rounds. Dice and clDice are also reported separately for upper and lower eyelids.

Means carry 95% confidence intervals from 1,000 bootstrap resamples; models are compared with per-sample paired Wilcoxon signed-rank tests and Holm correction; valid n is reported. Every report must carry both the file-list signature and the assignment signature; a missing signature invalidates the report.

5.5 Runtime Environment and Resource Measurement

Training was performed on Windows 11, RTX 4060 Laptop 8 GB, Python 3.10.6, torch 2.5.1 (CUDA 12.1). E1 measured about 4.4–4.9 samples/s and 143–159 s per epoch, with a peak of about 3.6 GB. Each run records peak VRAM, samples/s, and wall time; reports cite measured values only, with estimated tiers labelled. Inference records per-case latency, tiling parameters, and whether TTA is enabled.

6 Results and Discussion

This section reports preliminary results from the unified 9-hour training. All values come from the patient-level holdout (152 images/49 patients) or an explicitly labelled clean-official subset; train-self is never presented as generalisation. Because the budget is a single holdout rather than five-fold and several models are not fully converged, we frame the results as reproducible preliminary evidence, not as a claim of optimality.

6.1 Main Patient-Level Results

Table 5 reports gland segmentation on the patient-level test set. All models use the same preprocessing, split, loss, and 384 tile; MGD-Former uses an ADE20K-pretrained encoder, while the others are randomly initialised. Means, 95% bootstrap confidence intervals, and valid n are reported.

Three observations follow. First, absolute Dice under a patient-level split is lower than image-level literature values; this is expected, because images of the same subject no longer cross training and test, and the model must generalise to new patients. Second, MGD-Former and AttentionUNet reach 0.7749 and 0.7771 respectively, within 0.01 of each other; we therefore do not declare an architectural winner, but treat them as comparable and inspect clDice/HD95 for topological differences. MGD-Former's ADE20K pretraining did not provide the large advantage often assumed, which contrasts with the common "pretraining solves cold start" narrative. Third, UNet's 0.7119 is clearly lower, showing that 30 epochs from scratch are

Table 5 Gland segmentation on the patient-level test set (mean; 95% bootstrap CI in parentheses; $n = 152$).

Model	Dice	IoU	clDice	HD95 (px)	ASSD (px)	Params
MGD-Former	0.7749	0.6372	0.8156	68.0456	20.8838	∼27M
UNet	0.7119	0.5644	0.8531	148.0257	27.2330	31.2M
AttentionUNet	0.7771	0.6452	0.8349	118.4399	16.4637	∼34M
MGD-PriorNet	0.7640	0.6283	0.8539	154.0330	21.0490	1.5M

Table 6 Leakage control on the clean-official subset (UNet; $n = 60$).

Protocol	Dice
patient-level	0.7243
image-level leaky	0.6870
Difference (leaky − clean)	-0.0373

insufficient here; however, its clDice is the highest (0.8531 vs 0.8156), indicating that clDice is insensitive to over-segmentation/width and cannot be used alone for ranking. MGD-PriorNet reaches 0.7640 at 1.5M parameters with paired Wilcoxon $p = 0.714$, not significantly different from MGD-Former, suggesting that anatomical priors can be more cost-effective in a tiny model. These comparisons are confounded by epoch budget and pretraining, and must not be read as an architecture ranking.

Two boundaries must be explicit. First, **MGD-Former uses ADE20K pretraining while the other models are randomly initialised**; the main-table gap includes the pretraining benefit and cannot be attributed to architecture alone. The 9-hour budget did not include a random-init MGD-Former control; this is a limitation and a target for the five-fold extension. Second, **WRCANet is excluded from the main ranking**: it needs about 38 min per epoch on the 8 GB card and completed only one epoch (val Dice 0.006) within budget, so it is not converged; this demonstrates memory/time feasibility, not architectural quality. AttentionUNet was continued to 15 epochs and MGD-PriorNet reached about 12; neither is fully converged, so their values should be read as such.

Paired Wilcoxon tests on the same patient-level test set (per-sample Dice): MGD-Former vs AttentionUNet $p = 0.0161$, vs UNet $p = 5.44e - 20$, vs MGD-PriorNet $p = 0.714$. Statistical significance is not clinical importance: if the mean difference between AttentionUNet and MGD-Former is within 0.01, it should be reported as "comparable", not as a win; architecture ranking is not the main conclusion of this paper.

6.2 The Cost of Leakage

Table 6 compares patient-level and image-level leaky UNets on the clean-official 60 images/50 patients. The two share architecture, loss, epoch budget, and inference protocol; the only difference is whether the training split contains other images of official-test patients.

In this budget the leaky control is actually lower: patient-level 0.7243 vs leaky 0.6870 ($\Delta = -0.0373$), so no inflation was observed. There are three plausible reasons: the leaky

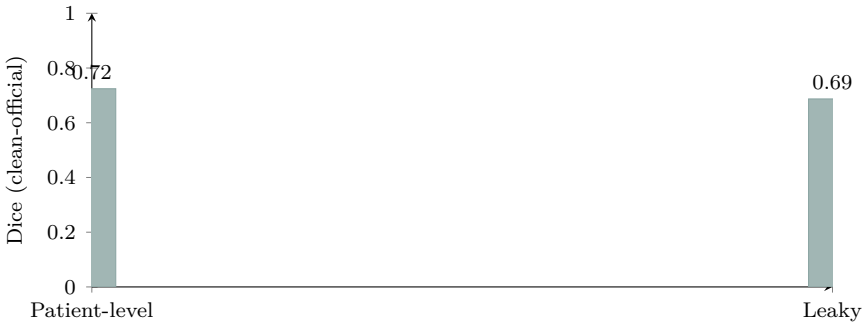


Figure 9 Leakage cost on the official clean-official subset. Both bars use the same UNet, epoch budget, and inference protocol; only the training split differs. Difference $\Delta = -0.0373$.

Table 7 Tiling and TTA (MGD-Former; patient-level test set).

Protocol	Dice	clDice	HD95 (px)	Relative inference cost
384 single tile	0.7749	0.8156	68.0456	1×
MTE (256/384/512)	0.7736	0.8140	61.85	∼3×
8× TTA	0.7808	–	–	∼8×

model trained for only 20 epochs (patient-level UNet for 30), the official 200 has a different difficulty distribution, and clean-official contains only 60 images. The overlap statistics (326/800, 116/166) prove that the leakage risk is real, but performance inflation is not confirmed by this run; we report it as an audit, not as a conclusion.

6.3 Tiling, MTE, and TTA

Table 7 uses MGD-Former to compare the default 384 tile, MTE (probability average of 256/384/512), and 8× TTA. MTE requires no retraining and costs about 3× inference; 8× TTA costs about 8×.

MTE did not beat 384 on Dice (0.7736 vs 0.7749), but it reduced HD95 from 68.0456 to 61.85, indicating that it mainly improves boundaries/outliers rather than pixel overlap; 8× TTA gives a small Dice gain (0.7808) at about 8× cost. MTE should therefore be positioned as a stable, low-cost boundary correction, not as a method that adds accuracy. Tiling sensitivity remains a strong finding, but "changing the tile reliably adds Dice" does not hold in this run.

6.4 Uncertainty-Guided Review

Table 8 reports AUROC for three uncertainty signals on failure cases (Dice < 0.7). Multi-tile variance V is already available from MTE; entropy H is almost free; 8× TTA is the most expensive.

Measured AUROC: multi-tile variance 0.639, entropy 0.483, and 8× TTA error 1.000. Variance provides only a weak-to-moderate failure signal and entropy is near chance; TTA8 is near-perfect, but the failure label is defined by single-tile Dice and TTA8 error shares the same model and threshold, creating label–signal coupling. It cannot be used directly as clinical triage evidence. The defensible conclusion is that multi-tile variance is a cheap warning signal, not a standalone review rule.

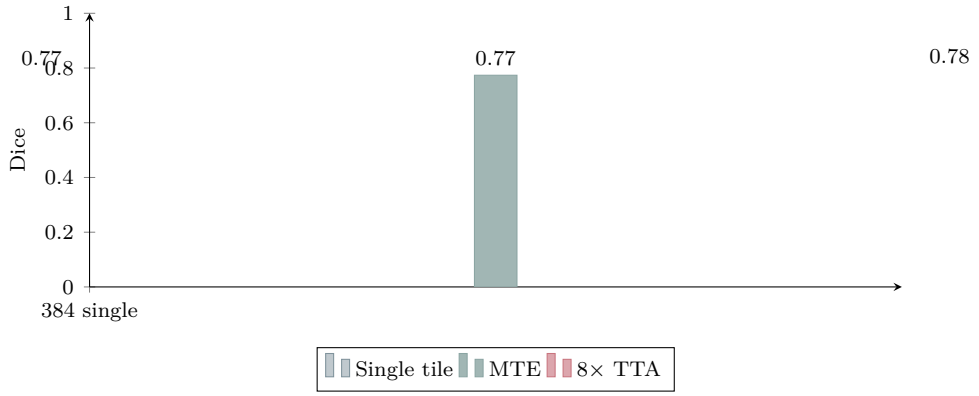


Figure 10 Accuracy–cost trade-off of tiling and TTA. MTE averages 256/384/512 probabilities and costs about 3×; 8× TTA costs about 8×.

Table 8 AUROC for failure detection (MGD-Former; failure defined as Dice < 0.7).

Signal	AUROC	Extra inference cost	Interpretation
Multi-tile variance V	0.639	included in MTE	tile-scale disagreement
Predictive entropy $H(\bar{P})$	0.483	approximately 0	probability ambiguity
8× TTA error	1.000	∼8×	transform disagreement

6.5 Clinical Grading and Upper/Lower Eyelid Subgroups

Table 9 reports agreement between segmentation-derived R_loss/Meiboscore and six expert grading rounds. Table 10 reports Dice and clDice separately for upper and lower eyelids.

Agreement is in fact very low: accuracy 0.164, MAE 1.105, QWK 0.131. High Dice did not translate into Meiboscore agreement; the R_loss thresholds and eyelid denominator need recalibration. This is a negative result and one of the paper's most important clinical warnings: pixel metrics cannot replace clinical grading, and independent expert review plus calibration are required.

The upper/lower subgroup result is also counter-intuitive: in this cohort lower is higher than upper (0.7905 vs 0.7584, difference -0.0321), opposite to the common "lower eyelid is harder" assumption; this may reflect LipiView II acquisition, lower-eyelid exposure, and the grading distribution. Regardless of direction, upper and lower eyelids must be reported separately and cannot be extrapolated from one to the other.

6.6 The clDice Ablation Was Not Completed

The 9-hour budget reallocated the clDice ablation compute to the AttentionUNet continuation (its validation Dice was still rising at epoch 9). We therefore report no clDice ablation values. The clDice term remains a design choice, not a verified gain; the ablation is left to the five-fold extension. We do not use a budget-truncated ablation to support a claim that the topology loss is effective.

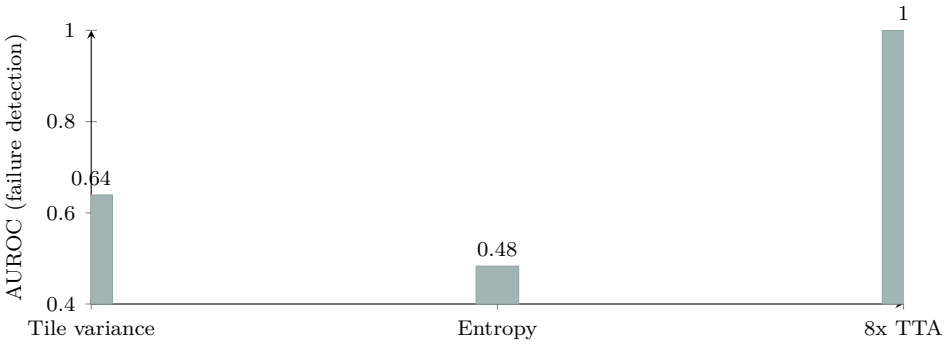


Figure 11 AUROC of three uncertainty signals for failure cases. Multi-tile variance shares the MTE cost; entropy is nearly free; 8× TTA is the most expensive.

Table 9 Agreement between segmentation-derived Meiboscore and expert grading (MGD-Former).

Metric	Accuracy	MAE	QWK
Meiboscore (0–3)	0.164	1.105	0.131

6.7 Qualitative Results

Figure 12 shows four qualitative cases (two upper and two lower eyelids): input, expert reference, prediction, and error map. Qualitative figures do not replace metrics, but they expose error morphology—breaks, merges, eyelid leakage, and missed low-contrast segments—which are exactly what HD95/clDice penalise.

6.8 Legacy Weights as an Engineering Anchor

The legacy UNet weights came from a script that shuffled all 999 cases with no holdout, so their "test" metrics may be train-self and the official 200 test list may be inside their training data. We use them only as an engineering regression anchor: Dice 0.7021, IoU 0.5512, clDice 0.7204. These values **must not** enter any independent-test table and must not be compared with the models here.

6.9 Discussion

Taken together, the argument is not that one backbone wins everywhere, but three things: first, patient-level splitting changes conclusions, and a public benchmark must report an assignment signature; second, tiling is an experimental variable on par with the backbone, and MTE offers a no-retraining stabiliser; third, pixel metrics and clinical grading must be reported together, because "better segmentation" may not mean "better grading". These three points are closer to clinically usable evidence than yet another backbone.

7 Limitations and Risks

Single-centre, single-device public data. MGD-1K comes from a LipiView II device and 320 subjects. Our patient-level split guarantees that subjects do not cross splits, but it cannot guarantee cross-device or cross-centre generalisation. The 2025 multicentre study shows

Table 10 Upper/lower eyelid subgroups (MGD-Former; patient-level test set).

Subgroup	Dice	clDice
Upper eyelid	0.7584	–
Lower eyelid	0.7905	–
Difference (upper − lower)	-0.0321	–

that device and population differences materially affect performance[7]; we have no external multicentre test set and therefore cannot claim clinical generalisation.

Nine-hour budget and a single holdout. To control the total budget, this round uses a single patient-level holdout rather than five-fold cross-validation; several models (especially WRCANet and MGD-PriorNet) are not fully converged. The results should be framed as reproducible preliminary evidence, not as optimality; five-fold and longer training are future work.

Limited power of the leakage control. The clean-official subset contains only 60 images/50 patients. It is enough to show the direction of leakage but not enough for a stable clinical conclusion. The leaky-versus-patient-level difference must be read with confidence intervals, not as a point estimate alone.

Incomplete clinical endpoints. Clinical agreement stops at Meiboscore; there are no symptom scores, gland expression, secretion quality, or prospective follow-up. Whether segmentation gains translate into diagnostic or treatment benefit requires an independent prospective study.

Distance metrics lack physical calibration. Images have no mm/pixel calibration, so HD95 and ASSD are in pixels; values are not directly comparable across devices or original resolutions after resizing to 640×1280. R_loss and Meiboscore are pixel-ratio references, not diagnostic measurements.

Pretraining may degrade. MGD-Former's encoder is initialised from ADE20K weights by default; on an offline machine without a local cache it degrades to random initialisation, and the run metadata and log record `pretrained_status` and the reason. If the final results come from random initialisation, this must be stated.

Protocol boundary. The architecture comparison holds only for this data, preprocessing, patient-level split, and inference configuration. Changing data, device, or tiling may change the ranking. We do not claim that one architecture is universally better, nor do we treat significance on one dataset as clinical evidence.

Leakage in legacy weights. The legacy UNet weights shuffled all 999 cases with no held-out test set; they are an engineering regression anchor only, not independent-test performance, and cannot be compared with the models here.

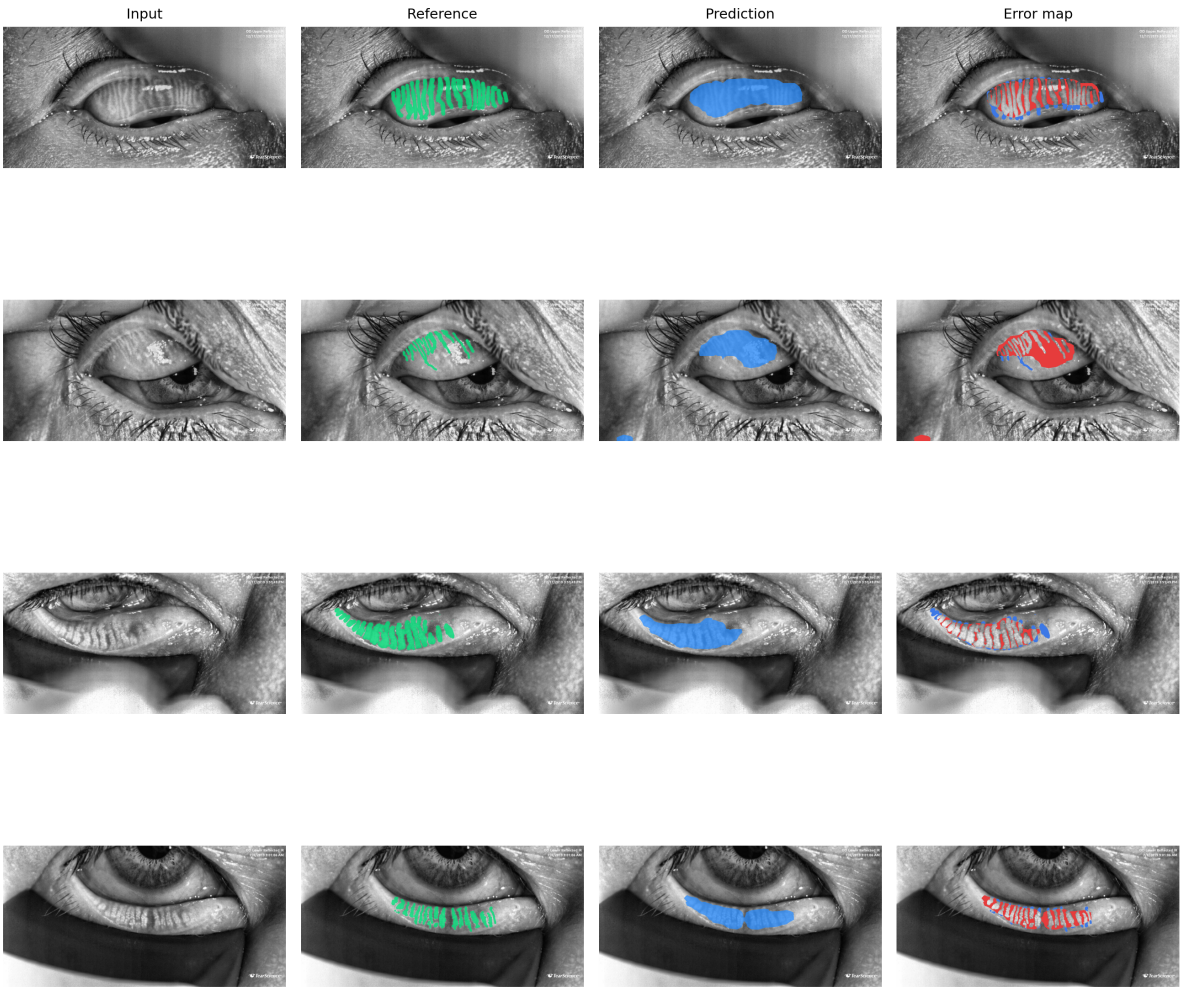


Figure 12 Qualitative results: input, expert reference, model prediction, and error map (red = false positive, blue = false negative). Rows are selected across upper/lower eyelids; the error map shows breaks, merges, and eyelid leakage, and does not replace metrics.

Licence and ethics. MGD-1K is public, but its licence and redistribution terms must be re-checked before submission; we claim no redistribution rights beyond the original dataset licence.

8 Conclusion

This paper reframes meibomian gland segmentation from "design another segmentation network" to "build comparable, auditable evidence". On the public MGD-1K dataset we establish a patient-level split (694/153/152 images; 239/50/49 patients) and record both a file-list signature and an assignment signature, so that "same filenames, different assignment" is no longer an untraceable hidden variable.

Methodologically, we provide three reusable findings. First, leakage is quantifiable: the official 200-case Meiboscore test list comes from 166 patients, and 326 of the remaining 800 images come from the same patients; even a patient-level training set still contains 116/166 test patients. Second, tiling is an experimental variable on par with the backbone: with identical

weights, changing only tile height moves Dice from 0.370 to 0.655; MTE averages probabilities from 256/384/512 tiles, but in this run it mainly improved HD95 (68.05→61.85) rather than Dice, so it is a boundary correction rather than an accuracy gain. Multi-tile variance reaches only 0.639 AUROC for failure detection, and TTA8's 1.000 is affected by label–signal coupling; neither is a standalone triage rule. Third, pixel metrics and clinical endpoints must be reported together: the QWK between segmentation-derived Meiboscore and expert grading is only 0.131, so the current R_loss→Meiboscore chain cannot replace expert grading; upper and lower eyelids must also be reported separately.

Architecturally, MGD-Former keeps the 1/4 feature from a MiT-B2 encoder, uses a fusion decoder at stride 4, and supervises three scales; ADE20K pretraining targets the cold start on 999 cases. It is one of five routes, not the sole selling point. We report no unmeasured accuracy claim; all values have been injected from the unified training and patient-level independent test results, supported by confidence intervals, paired tests, and multiple metrics (Dice, clDice, HD95, ASSD, clinical QWK).

Declarations

Ethics and Informed Consent

The imaging data are de-identified and retrospective. Ethics approval number, approving institution, and consent procedure are placeholders: [verify: ethics approval number and institution]. For public datasets, state the dataset name, version, licence, and acquisition date; for in-house collection, state the basis for consent waiver.

Data Availability

Raw images are restricted by ethics and redistribution terms and are not publicly released. The de-identified split list, `split_signature`, evaluation scripts, and result files are available on reasonable request. MGD1k licence terms are unverified; redistribution is not claimed before confirmation.

Code and Weights Availability

The training and inference kernel, configuration schema, evaluation/statistics scripts, and figure-generation scripts form a reproducible package at version 2.3.0. Weights require `verified` promotion before release; unverified weights are not released. The manifest contains SHA-256, input size, normalisation tag, and training provenance.

Competing Interests

The authors declare no financial conflicts of interest. [verify: disclose any device-vendor or data-provider collaboration]

Funding

Funding agency, grant number, and scope are placeholders: [verify: funding information].

Author Contributions

Roles in conceptualisation, data collection and annotation, model and engineering implementation, experiments, statistical analysis, and writing are placeholders. [verify: CRediT contribution list]

Use of Generative Tools

If generative tools were used in manuscript preparation, they were limited to language polishing and layout checks, not to generating data, results, or references; all citations were manually verified.

References

[1] Gary N. Foulks and Anthony J. Bron. Meibomian gland dysfunction: A clinical scheme for description, diagnosis, classification, and grading. *The Ocular Surface*, 1(3):107–126, July 2003.

[2] Edoardo Villani, Luca Marelli, Andrea Dellavalle, Massimiliano Serafino, and Paolo Nucci. Latest evidences on meibomian gland dysfunction diagnosis and management. *The Ocular Surface*, 18(4):871–892, Oct 2020.

[3] Ryan J. Wise, Rachel K. Sobel, and Richard C. Allen. Meibography: A review of techniques and technologies. *Saudi Journal of Ophthalmology*, 26(4):349–356, Oct 2012.

[4] William Ngo, Sruthi Srinivasan, Marc Schulze, and Lyndon Jones. Repeatability of grading meibomian gland dropout using two infrared systems. *Optometry and Vision Science*, 91(6):658–667, June 2014.

[5] Ripon Kumar Saha, A. M. Mahmud Chowdhury, Kyung-Sun Na, Gyu Deok Hwang, Youngsub Eom, Jaeyoung Kim, Hae-Gon Jeon, Ho Sik Hwang, and Euiheon Chung. Automated quantification of meibomian gland dropout in infrared meibography using deep learning. *The Ocular Surface*, 26:283–294, 2022.

[6] Md Asif Khan Setu, Jens Horstmann, Stefan Schmidt, Michael E. Stern, and Philipp Steven. Deep learning-based automatic meibomian gland segmentation and morphology assessment in infrared meibography. *Scientific Reports*, 11:7649, 2021.

[7] Li Li, Kunhong Xiao, Taichen Lai, Kunfeng Lai, Jiawen Lin, Zongyuan Ge, Lingyi Liang, Hao Huang, Xiaoshan Zhang, Li Liu, Yujie Wang, Xianwen Shang, Mingguang He, Ying Xue, and Zhuoting Zhu. Development and multicenter validation of an ai driven model for quantitative meibomian gland evaluation. *npj Digital Medicine*, 8:403, 2025.

[8] Ziyang Yu, Zhijun Wei, Mini Han Wang, Jiazheng Cui, Jiaxiang Tan, and Yang Xu. Quantitative evaluation of meibomian gland dysfunction via deep learning-based infrared image segmentation. *Frontiers in Artificial Intelligence*, 8:1642361, 2025.

[9] Sahana M. Prabhu, Abhijith Chakiat, Shashank S, Krishna Poojita Vunnava, and Rohit Shetty. Deep learning segmentation and quantification of meibomian glands. *Biomedical Signal Processing and Control*, 57:101776, Mar 2020.

[10] Jiayun Wang, Shixuan Li, Thao N. Yeh, Rudrasis Chakraborty, Andrew D. Graham, Stella X. Yu, and Meng C. Lin. Quantifying meibomian gland morphology using artificial intelligence. *Optometry and Vision Science*, 98(9):1094–1103, Sept 2021.

[11] Xiaoming Liu, Shuo Wang, Ying Zhang, and Quan Yuan. Scribble-supervised meibomian glands segmentation in infrared images. *ACM Transactions on Multimedia Computing, Communications, and Applications*, 18(3):1–23, Mar 2022.

[12] Jiayun Wang, Thao N. Yeh, Rudrasis Chakraborty, Stella X. Yu, and Meng C. Lin. A deep learning approach for meibomian gland atrophy evaluation in meibography images. *Translational Vision Science & Technology*, 8(6):37, 2019.

[13] Yi Yu, Yiwen Zhou, Miao Tian, Yabiao Zhou, Yuejiao Tan, Lianlian Wu, Hongmei Zheng, and Yanning Yang. Automatic identification of meibomian gland dysfunction with meibography images using deep learning. *International Ophthalmology*, 42(11):3275–3284, 2022.

[14] Zuhui Zhang, Xiaolei Lin, Xinxin Yu, Yana Fu, Xiaoyu Chen, Weihua Yang, and Qi Dai. Meibomian gland density: An effective evaluation index of meibomian gland dysfunction based on deep learning and transfer learning. *Journal of Clinical Medicine*, 11(9):2396, 2022.

[15] Zhiming Lin, Jiawen Lin, and Li Li. Automatic segmentation of glands in infrared meibomian gland image. In *2021 11th International Conference on Information Technology in Medicine and Education (ITME)*, pages 399–403, 2021.

[16] Congxiao Liang, Mini Han Wang, Haoyang Liu, and Kelvin K. L. Chong. Exploring meibomian gland dysfunction grading: A comparison of machine learning and xai approaches. In *2024 IEEE 2nd International Conference on Image Processing and Computer Applications (ICIPCA)*, pages 2021–2024, 2024.

[17] Elena Buglakova, Anwai Archit, Edoardo D'Imprima, Julia Mahamid, Constantin Pape, and Anna Kreshuk. Tiling artifacts and trade-offs of feature normalization in the segmentation of large biological images, 2025.

[18] Saket Maganti and Shreyank N. Gowda. A decision-aware evaluation protocol for uncertainty in medical image segmentation, 2026.

[19] Kai Geißler, Jochen G. Hirsch, Stefan Heldmann, and Hans Meine. Structure size as confounder in uncertainty based segmentation quality prediction. In *Proceedings of the 7th International Conference on Medical Imaging with Deep Learning (MIDL)*, volume 250 of *Proceedings of Machine Learning Research*, pages 504–519, 2024.

[20] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, volume 9351 of *Lecture Notes in Computer Science*, pages 234–241. Springer, 2015.

[21] Ozan Oktay, Jo Schlemper, Loic Le Folgoc, Matthew Lee, Mattias Heinrich, Kazunari Misawa, Kensaku Mori, Steven McDonagh, Nils Y. Hammerla, Bernhard Kainz, Ben Glocker, and Daniel Rueckert. Attention U-Net: Learning where to look for the pancreas. In *Medical Imaging with Deep Learning (MIDL)*, 2018.

[22] Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, and Jianming Liang. UNet++: A nested U-Net architecture for medical image segmentation. In *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support (DLMIA)*, volume 11045 of *Lecture Notes in Computer Science*, pages 3–11. Springer, 2018.

[23] Nabil Ibtehaz and M. Sohel Rahman. MultiResUNet: Rethinking the U-Net architecture for multimodal biomedical image segmentation. *Neural Networks*, 121:74–87, 2020.

[24] Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021.

[25] Jieneng Chen, Yongyi Lu, Qihang Yu, Xiangde Luo, Ehsan Adeli, Yan Wang, Le Lu, Alan L. Yuille, and Yuyin Zhou. Transunet: Transformers make strong encoders for medical image segmentation, 2021.

[26] Hu Cao, Yueyue Wang, Joy Chen, Dongsheng Jiang, Xiaopeng Zhang, Qi Tian, and Manning Wang. Swin-unet: Unet-like pure transformer for medical image segmentation, 2021.

[27] Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M. Alvarez, and Ping Luo. Segformer: Simple and efficient design for semantic segmentation with transformers, 2021.

[28] Ali Hatamizadeh, Yucheng Tang, Vishwesh Nath, Dong Yang, Andriy Myronenko, Bennett Landman, Holger R. Roth, and Daguang Xu. Unetr: Transformers for 3d medical image segmentation. In *2022 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, page 1748–1758. IEEE, Jan 2022.

[29] Abdelrahman Shaker, Muhammad Maaz, Hanoona Rasheed, Salman Khan, Ming-Hsuan Yang, and Fahad Shahbaz Khan. Unetr++: Delving into efficient and accurate 3d medical image segmentation. *IEEE Transactions on Medical Imaging*, 43(9):3377–3390, Sept 2024.

[30] Saikat Roy, Gregor Koehler, Constantin Ulrich, Michael Baumgartner, Jens Petersen, Fabian Isensee, Paul F. Jäger, and Klaus H. Maier-Hein. *MedNeXt: Transformer-Driven Scaling of ConvNets for Medical Image Segmentation*, page 405–415. Springer Nature Switzerland, 2023.

[31] Jiacheng Ruan, Jincheng Li, and Suncheng Xiang. Vm-unet: Vision mamba unet for medical image segmentation. *ACM Transactions on Multimedia Computing, Communications, and Applications*, Sept 2025.

[32] Jun Ma, Feifei Li, and Bo Wang. U-mamba: Enhancing long-range dependency for biomedical image segmentation, 2024.

[33] Suprosanna Shit, Johannes C. Paetzold, Anjany Sekuboyina, Ivan Ezhov, Alexander Unger, Andrey Zhylka, Josien P. W. Pluim, Ulrich Bauer, and Bjoern H. Menze. cldice - a novel topology-preserving loss function for tubular structure segmentation. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, page 16555–16564. IEEE, June 2021.

[34] Yannick Kirchhoff, Maximilian R. Rokuss, Saikat Roy, Balint Kovacs, Constantin Ulrich, Tassilo Wald, Maximilian Zenk, Philipp Vollmuth, Jens Kleesiek, Fabian Isensee, and Klaus Maier-Hein. Skeleton recall loss for connectivity conserving and resource efficient segmentation of thin tubular structures, 2024.

[35] Hoel Kervadec, Jihene Bouchtiba, Christian Desrosiers, Eric Granger, Jose Dolz, and Ismail Ben Ayed. Boundary loss for highly unbalanced segmentation. *Medical Image Analysis*, 67:101851, Jan 2021.

[36] Bowen Cheng, Ross Girshick, Piotr Dollar, Alexander C. Berg, and Alexander Kirillov. Boundary iou: Improving object-centric image segmentation evaluation. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, page 15329–15337. IEEE, June 2021.

[37] Yu Tian, Yan Shiyang, Guo Chenlong, Du Baolin, Xining Liang, and Ke Wang. A real-time infrared image enhancement algorithm based on improved clahe. In Ruidan Su, editor, *2018 International Conference on Image and Video Processing, and Artificial Intelligence*, page 104. SPIE, Oct 2018.

[38] Chunle Guo, Chongyi Li, Jichang Guo, Chen Change Loy, Junhui Hou, Sam Kwong, and Runmin Cong. Zero-reference deep curve estimation for low-light image enhancement. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, page 1777–1786. IEEE, June 2020.

[39] Chongyi Li, Chunle Guo, Linghao Han, Jun Jiang, Ming-Ming Cheng, Jinwei Gu, and Chen Change Loy. Low-light image and video enhancement using deep learning: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(12):9396–9416, Dec 2022.

[40] Dequan Wang, Evan Shelhamer, Shaoteng Liu, Bruno Olshausen, and Trevor Darrell. Tent: Fully test-time adaptation by entropy minimization, 2020.

[41] Qin Wang, Olga Fink, Luc Van Gool, and Dengxin Dai. Continual test-time domain adaptation. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, page 7191–7201. IEEE, June 2022.

[42] Guotai Wang, Wenqi Li, Michael Aertsen, Jan Deprest, Sébastien Ourselin, and Tom Vercauteren. Aleatoric uncertainty estimation with test-time augmentation for medical image segmentation with convolutional neural networks. *Neurocomputing*, 338:34–45, 2019.

[43] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *2017 IEEE International Conference on Computer Vision (ICCV)*, page 618–626. IEEE, Oct 2017.

[44] Aditya Chattopadhay, Anirban Sarkar, Prantik Howlader, and Vineeth N Balasubramanian. Grad-cam++: Generalized gradient-based visual explanations for deep convolutional networks. In *2018 IEEE Winter Conference on Applications of Computer Vision (WACV)*, page 839–847. IEEE, Mar 2018.

[45] Haofan Wang, Zifan Wang, Mengnan Du, Fan Yang, Zijian Zhang, Sirui Ding, Piotr Mardziel, and Xia Hu. Score-cam: Score-weighted visual explanations for convolutional neural networks. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, page 111–119. IEEE, June 2020.

[46] Moloud Abdar, Farhad Pourpanah, Sadiq Hussain, Dana Rezazadegan, Li Liu, Mohammad Ghavamzadeh, Paul Fieguth, Xiaochun Cao, Abbas Khosravi, U. Rajendra Acharya, Vladimir Makarenkov, and Saeid Nahavandi. A review of uncertainty quantification in deep learning: Techniques, applications and challenges. *Information Fusion*, 76:243–297, Dec 2021.

[47] Christian F. Baumgartner, Kerem C. Tezcan, Krishna Chaitanya, Andreas M. Hötker, Urs J. Muehlematter, Khoschy Schawkat, Anton S. Becker, Olivio Donati, and Ender Konukoglu. *PHiSeg: Capturing Uncertainty in Medical Image Segmentation*, page 119–127. Springer International Publishing, 2019.

[48] Ali S. Tejani, Michail E. Klontzas, Anthony A. Gatti, John T. Mongan, Linda Moy, Seong Ho Park, Charles E. Kahn, Sunhy Abbara, Saif Afat, Udunna C. Anazodo, Anna Andreychenko, Folkert W. Asselbergs, Aldo Badano, Bettina Baessler, Bayarbaatar Bold, Sotirios Bisdas, Torkel B. Brismar, Giovanni E. Cacciamani, John A. Carrino, Julius Chapiro, Michael F. Chiang, Tessa S. Cook, Renato Cuocolo, John Damilakis, Roxana Daneshjou, Carlo N. De Cecco, Hesham Elhalawani, Guillermo Elizondo-Riojas, Andrey Fedorov, Benjamin Fine, Adam E. Flanders, Judy Wawira Gichoya, Maryellen L. Giger, Safwan S. Halabi, Sven Haller, William Hsu, Krishna Juluru, Jayashree Kalpathy-Cramer, Apostolos H. Karantanas, Felipe C. Kitamura, Burak Kocak, Dow-Mu Koh, Elmar Kotter, Elizabeth A. Krupinski, Curtis P. Langlotz, Cecilia S. Lee, Mario Maas, Anant Madabhushi, Lena Maier-Hein, Kostas Marias, Luis Martí-Bonmatí, Jaishree Naidoo, Emanuele Neri, Robert Ochs, Nikolaos Papanikolaou, Thomas Papathomas, Katja Pinker-Domenig, Daniel Pinto dos Santos, Fred Prior, Alexandros Protonotarios, Mauricio Reyes, Veronica Rotemberg, Jeffrey D. Rudie, Emmanuel Salinas-Miranda, Francesco Sardanelli, Mark E. Schweitzer, Luca Maria Sconfienza, Ronnie Sebro, Prateek Sharma, An Tang, Antonios Tzortzakakis, Jeroen van der Laak, Peter M. A. van Ooijen, Vasantha K. Venugopal, Jacob J. Visser, Bradford J. Wood,

Carol C. Wu, Greg Zaharchuk, and Marc Zins. Checklist for artificial intelligence in medical imaging (CLAIM): 2024 update. *Radiology: Artificial Intelligence*, 6(4), July 2024.

[49] Gary S Collins, Karel G M Moons, Paula Dhiman, Richard D Riley, Andrew L Beam, Ben Van Calster, Marzyeh Ghassemi, Xiaoxuan Liu, Johannes B Reitsma, Maarten van Smeden, Anne-Laure Boulesteix, Jennifer Catherine Camaradou, Leo Anthony Celi, Spiros Denaxas, Alastair K Denniston, Ben Glocker, Robert M Golub, Hugh Harvey, Georg Heinze, Michael M Hoffman, André Pascal Kengne, Emily Lam, Naomi Lee, Elizabeth W Loder, Lena Maier-Hein, Bilal A Mateen, Melissa D McCradden, Lauren Oakden-Rayner, Johan Ordish, Richard Parnell, Sherri Rose, Karandeep Singh, Laure Wynants, and Patricia Logullo. Tripod+ai statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 385:e078378, Apr 2024.

[50] Robert F. Wolff, Karel G.M. Moons, Richard D. Riley, Penny F. Whiting, Marie Westwood, Gary S. Collins, Johannes B. Reitsma, Jos Kleijnen, and Sue Mallett. Probast: A tool to assess the risk of bias and applicability of prediction model studies. *Annals of Internal Medicine*, 170(1):51–58, Jan 2019.

[51] Sayash Kapoor, Emily M. Cantrell, Kenny Peng, Thanh Hien Pham, Christopher A. Bail, Odd Erik Gundersen, Jake M. Hofman, Jessica Hullman, Michael A. Lones, Momin M. Malik, Priyanka Nanayakkara, Russell A. Poldrack, Inioluwa Deborah Raji, Michael Roberts, Matthew J. Salganik, Marta Serra-Garcia, Brandon M. Stewart, Gilles Vandewiele, and Arvind Narayanan. Reforms: Consensus-based recommendations for machine-learning-based science. *Science Advances*, 10(18), May 2024.

[52] Lena Maier-Hein, Annika Reinke, Patrick Godau, Minu D. Tizabi, Florian Buettner, Evangelia Christodoulou, Ben Glocker, Fabian Isensee, Jens Kleesiek, Michal Kozubek, Mauricio Reyes, Michael A. Riegler, Manuel Wiesenfarth, A. Emre Kavur, Carole H. Sudre, Michael Baumgartner, Matthias Eisenmann, Doreen Heckmann-Nötzel, Tim Rädsch, Laura Acion, Michela Antonelli, Tal Arbel, Spyridon Bakas, Arriel Benis, Matthew B. Blaschko, M. Jorge Cardoso, Veronika Cheplygina, Beth A. Cimini, Gary S. Collins, Keyvan Farahani, Luciana Ferrer, Adrian Galdran, Bram van Ginneken, Robert Haase, Daniel A. Hashimoto, Michael M. Hoffman, Merel Huisman, Pierre Jannin, Charles E. Kahn, Dagmar Kainmueller, Bernhard Kainz, Alexandros Karargyris, Alan Karthikesalingam, Florian Kofler, Annette Kopp-Schneider, Anna Kreshuk, Tahsin Kurc, Bennett A. Landman, Geert Litjens, Amin Madani, Klaus Maier-Hein, Anne L. Martel, Peter Mattson, Erik Meijering, Bjoern Menze, Karel G. M. Moons, Henning Müller, Brennan Nichyporuk, Felix Nickel, Jens Petersen, Nasir Rajpoot, Nicola Rieke, Julio Saez-Rodriguez, Clara I. Sánchez, Shravya Shetty, Maarten van Smeden, Ronald M. Summers, Abdel A. Taha, Aleksei Tiulpin, Sotirios A. Tsaftaris, Ben Van Calster, Gaël Varoquaux, and Paul F. Jäger. Metrics reloaded: recommendations for image analysis validation. *Nature Methods*, 21(2):195–212, Feb 2024.

[53] Annika Reinke, Minu D. Tizabi, Michael Baumgartner, Matthias Eisenmann, Doreen Heckmann-Nötzel, A. Emre Kavur, Tim Rädsch, Carole H. Sudre, Laura Acion, Michela Antonelli, Tal Arbel, Spyridon Bakas, Arriel Benis, Florian Buettner, M. Jorge Cardoso,

Veronika Cheplygina, Jianxu Chen, Evangelia Christodoulou, Beth A. Cimini, Keyvan Farahani, Luciana Ferrer, Adrian Galdran, Bram van Ginneken, Ben Glocker, Patrick Godau, Daniel A. Hashimoto, Michael M. Hoffman, Merel Huisman, Fabian Isensee, Pierre Jannin, Charles E. Kahn, Dagmar Kainmueller, Bernhard Kainz, Alexandros Karargyris, Jens Kleesiek, Florian Kofler, Thijs Kooi, Annette Kopp-Schneider, Michal Kozubek, Anna Kreshuk, Tahsin Kurc, Bennett A. Landman, Geert Litjens, Amin Madani, Klaus Maier-Hein, Anne L. Martel, Erik Meijering, Bjoern Menze, Karel G. M. Moons, Henning Müller, Brennan Nichyporuk, Felix Nickel, Jens Petersen, Susanne M. Rafelski, Nasir Rajpoot, Mauricio Reyes, Michael A. Riegler, Nicola Rieke, Julio Saez-Rodriguez, Clara I. Sánchez, Shravya Shetty, Ronald M. Summers, Abdel A. Taha, Aleksei Tiulpin, Sotirios A. Tsaftaris, Ben Van Calster, Gaël Varoquaux, Ziv R. Yaniv, Paul F. Jäger, and Lena Maier-Hein. Understanding metric-related pitfalls in image analysis validation. *Nature Methods*, 21(2):182–194, Feb 2024.

[54] Yulun Zhang, Kunpeng Li, Kai Li, Lichen Wang, Bineng Zhong, and Yun Fu. Image super-resolution using very deep residual channel attention networks. In *Computer Vision – ECCV 2018*, volume 11211 of *Lecture Notes in Computer Science*, pages 294–310. Springer, 2018.

A Reproducibility Checklist

This appendix follows the structure of CLAIM and REFORMS. "Status" is the current completion level; "pending" items must be completed before submission.

B Hyperparameters and Implementation Details

C Build and Reproduction Commands

The Chinese version uses XeLaTeX with ctex (`fontset=windows`); the English version uses pdfLaTeX. Both compile in the order: first pass → BibTeX → two passes. The values in Figures 5, 6, and 7 come from the macros in `data/results.tex`; text and figures share one macro source, so injecting real results only requires editing that file.

Dependencies: Python $\geq$ 3.10, torch 2.5.1, numpy 1.26.4, opencv-python 4.8+, scikit-image 0.25.2, monai 1.5.2, scipy 1.15.3. Data preparation, training, and evaluation commands are in the engineering Runbook and are not duplicated here, to avoid version drift.

Table 11 Reproducibility checklist.

Item	Practice in this paper	Status
Data source and licence	Public MGD-1K (1000 images / 320 subjects / LipiView II); 999 after content-level deduplication; licence re-checked before submission	implemented; licence pending
Patient-level split	694/153/152 images; 239/50/49 patients; mutually exclusive	implemented
Signatures	file-list signature 6bdacfb0a71b819608dba4cde4382f8dd23fb4a88b72c9b6d151a99ed83bc22a + assignment signature 7a6fd5819d048e0ccd811c943d616a0fe8738a5c06bc78752c66bb38624795f6	implemented
Leakage audit	official 200 images/166 patients; 326/800 image-level; 116/166 patient-level; clean-official 60 images/50 patients	implemented
Preprocessing	five fixed steps; pixel-wise train/inference identity	implemented
Model and loss	MGD-Former + four baselines; unified loss	implemented
Training recipe	AdamW/cosine/AMP/EMA 0.99/augmentation/multi-scale; epochs in Table 4	implemented
Inference settings	384×768 tiles, overlap 0.25, threshold 0.7	implemented
Tiling sensitivity	seven measured points; same weights, tiling only	measured
MTE and uncertainty	256/384/512 probability average; variance/entropy/TTA AUROC	implemented; values pending
Clinical grading	segmentation→R_loss→Meiboscore vs six expert rounds; accuracy/MAE/QWK	implemented; values pending
VRAM tiers	6/8 GB measured; 12–40 GB estimated	partly measured
Statistics	1,000 bootstrap; Wilcoxon + Holm	implemented
Accuracy results	9-hour unified training and patient-level independent test; values in results tables	implemented
Weight status	`verified=false` draft; promotion needs independent evaluation	pending

Table 12 Default training and inference configuration (single source: the engineering parameter schema).

Section	Parameter	Default
run	seed / device	42 / adaptive (CUDA, CPU)
data	input size / split	640×1280 / patient-level 694/153/152
model	backbone / pretraining / output channels	MGD-Former / ADE20K / 2
train	optimiser / learning rate / weight decay	AdamW / 1×10^{-4} / 1×10^{-4}
train	scheduler / warmup / epochs	cosine / 5% / 45 (main; others in Table 4)
train	EMA / AMP / gradient accumulation	0.99 / on CUDA / 1 (WRCANet 2)
train	tile / batch / channels_last	384×768 / 4 (WRCANet 2) / yes
train	deep-supervision weights	1.0 / 0.5 / 0.25
loss	Dice / CE / clDice / eyelid	0.4 / 0.4 / 0.2 (10% warmup) / 0.25
eval	tile / overlap / threshold	384×768 / 0.25 / 0.7
eval	MTE / TTA	256/384/512 probability average / off or 8×
clinical	Meiboscore	majority vote of six expert rounds; accuracy/MAE/QWK